

Arquitectura de Computadores

Clase: Camino de Datos Monociclo

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GRUPO SIRIUS



Universidad Tecnológica
de Pereira

De alto nivel a bajo nivel y máquina

Lenguaje C

`f = (g + h) - (i + j);` Sentencias

↓
`f → x8`
`g → x9`
`h → x18`
`i → x19`
`j → x20`

32 Registros de 32 bits
(x0, x1, ..., x31)

Lenguaje Ensamblador RISC-V

`add x6, x9, x18`
`add x7, x19, x20`
`sub x8, x6, x7`

Instrucciones

De alto nivel a bajo nivel y máquina

Lenguaje Ensamblador RISC-V

```
add x6, x9, x18
```

```
add x7, x19, x20
```

```
sub x8, x6, x7
```

Lenguaje de Máquina RISC-V

0x01248333	0000	0001	0010	0100	1000	0011	0011	0011
------------	------	------	------	------	------	------	------	------

0x014983B3	0000	0001	0100	1001	1000	0011	1011	0011
------------	------	------	------	------	------	------	------	------

0x40730433	0100	0000	0111	0011	0000	0100	0011	0011
------------	------	------	------	------	------	------	------	------

Instrucciones RISC-V

	31		25 24				20 19				15 14				12 11		7 6		0							
R			Funct7				rs2				rs1				Funct3				rd				OpCode			
I					Imm						rs1				Funct3				rd				OpCode			
S B			Imm				rs2				rs1				Funct3				Imm				OpCode			
U J							Imm												rd				OpCode			

Instrucciones RV32I

Tipo R

	OpCode	Funct3	Funct7
add	0110011	000	00000000
sub	0110011	000	01000000
sll	0110011	001	00000000
slt	0110011	010	00000000
sltu	0110011	011	00000000
xor	0110011	100	00000000
srl	0110011	101	00000000
sra	0110011	101	01000000
or	0110011	110	00000000
and	0110011	111	00000000

Tipo U J

	OpCode
jal	1101111
lui	0110111
auipc	0010111

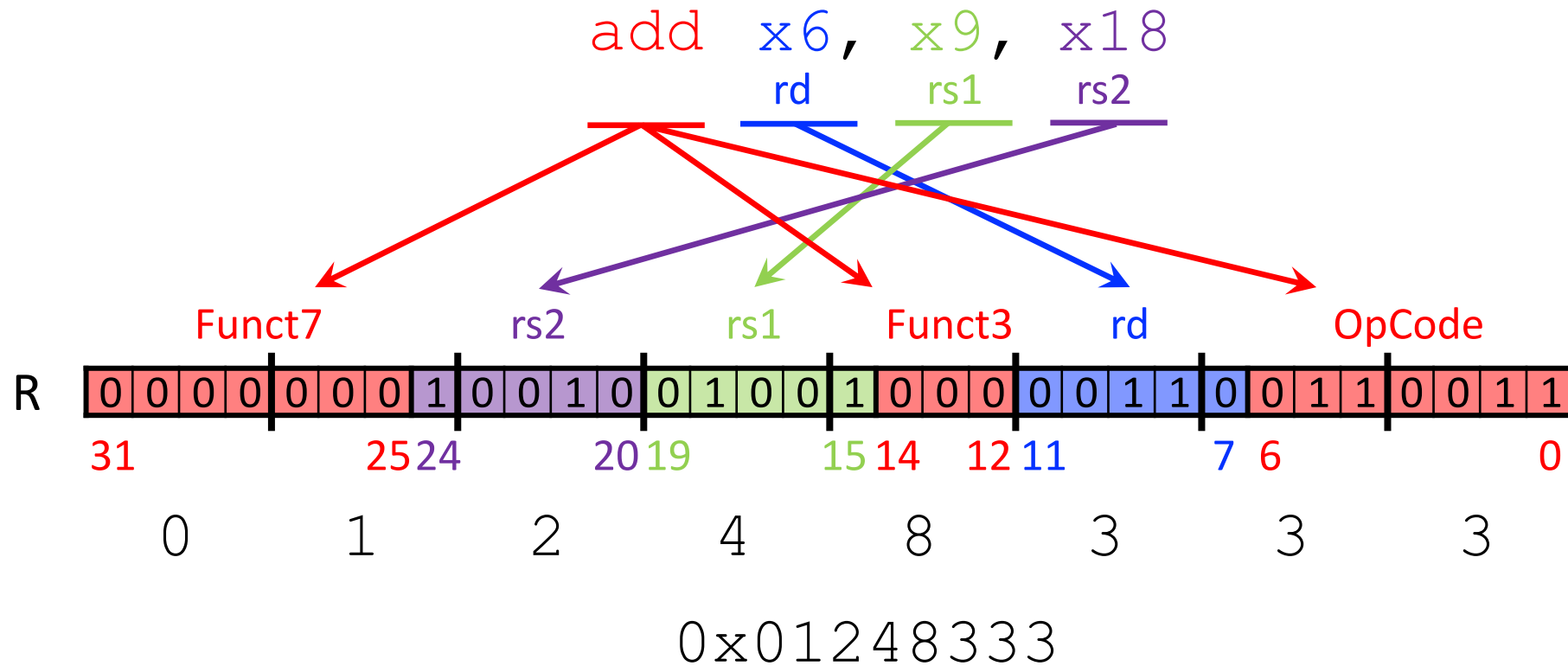
Tipo I

	OpCode	Funct3
addi	0010011	000
slli	0010011	001
slti	0010011	010
sltiu	0010011	011
xori	0010011	100
srli	0010011	101
srai	0010011	101
ori	0010011	110
andi	0010011	111
lb	0000011	000
lh	0000011	001
lw	0000011	010
lbu	0000011	100
lhu	0000011	101
jalr	1100111	000

Tipo S B

	OpCode	Funct3
beq	1100011	000
bne	1100011	001
blt	1100011	100
bge	1100011	101
bltu	1100011	110
bgeu	1100011	111
sb	0100011	000
sh	0100011	001
sw	0100011	010

Instrucciones Tipo R



Procesador
RISC-V
Monociclo

	31	25 24				20 19	15 14	12 11	7 6	0
R		Funct7			rs2		rs1	Funct3	rd	OpCode
I		Imm[11:0]					rs1	Funct3	rd	OpCode
S		Imm[11:5]				rs2	rs1	Funct3	Imm[4:0]	OpCode
B		Imm[12,10:5]				rs2	rs1	Funct3	Imm[4:1,11]	OpCode
U				Imm[31:12]					rd	OpCode
J			Imm[20,10:1,11,19:12]						rd	OpCode

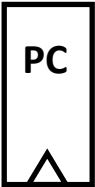
Procesador
RISC-V
Monociclo

Program
Counter



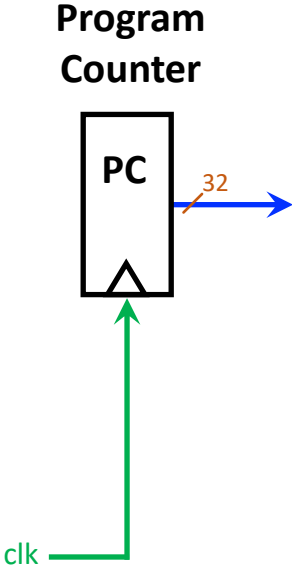
	31		25	24		20	19		15	14		12	11		7	6		0
R																		
I																		
S																		
B																		
U																		
J																		

Program
Counter

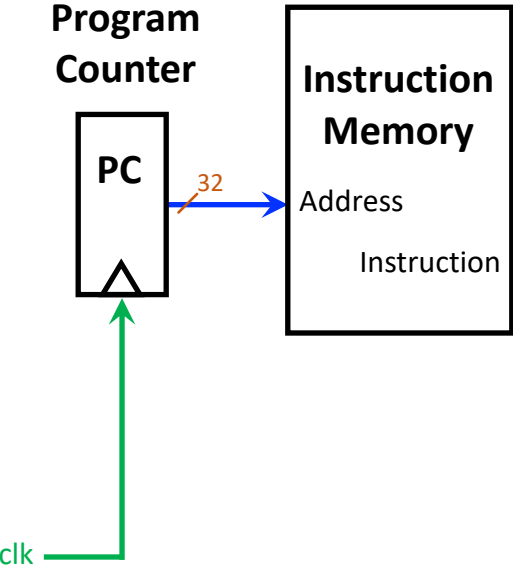


clk

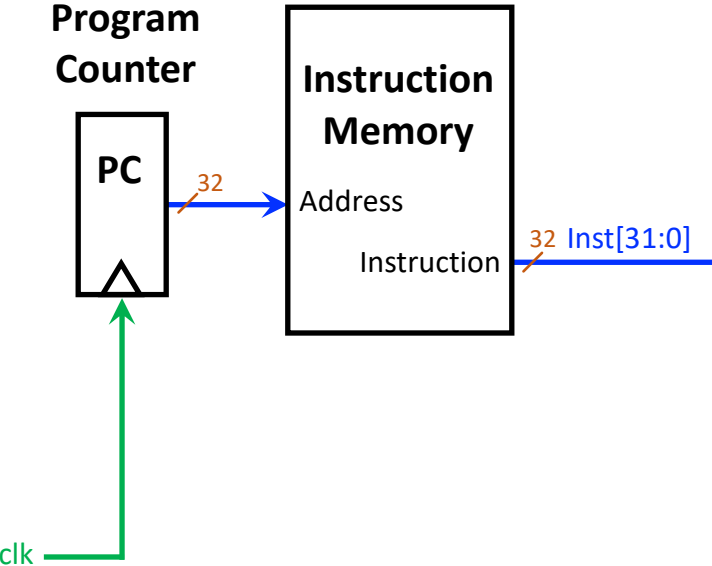
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R		Funct7				rs2		rs1		OpCode
I		Imm[11:0]					rs1	Funct3	rd	
S		Imm[11:5]		rs2		rs1		Funct3	Imm[4:0]	OpCode
B		Imm[12,10:5]				rs2		rs1	Funct3	Imm[4:1,11]
U						Imm[31:12]				rd
J						Imm[20,10:1,11,19:12]				rd



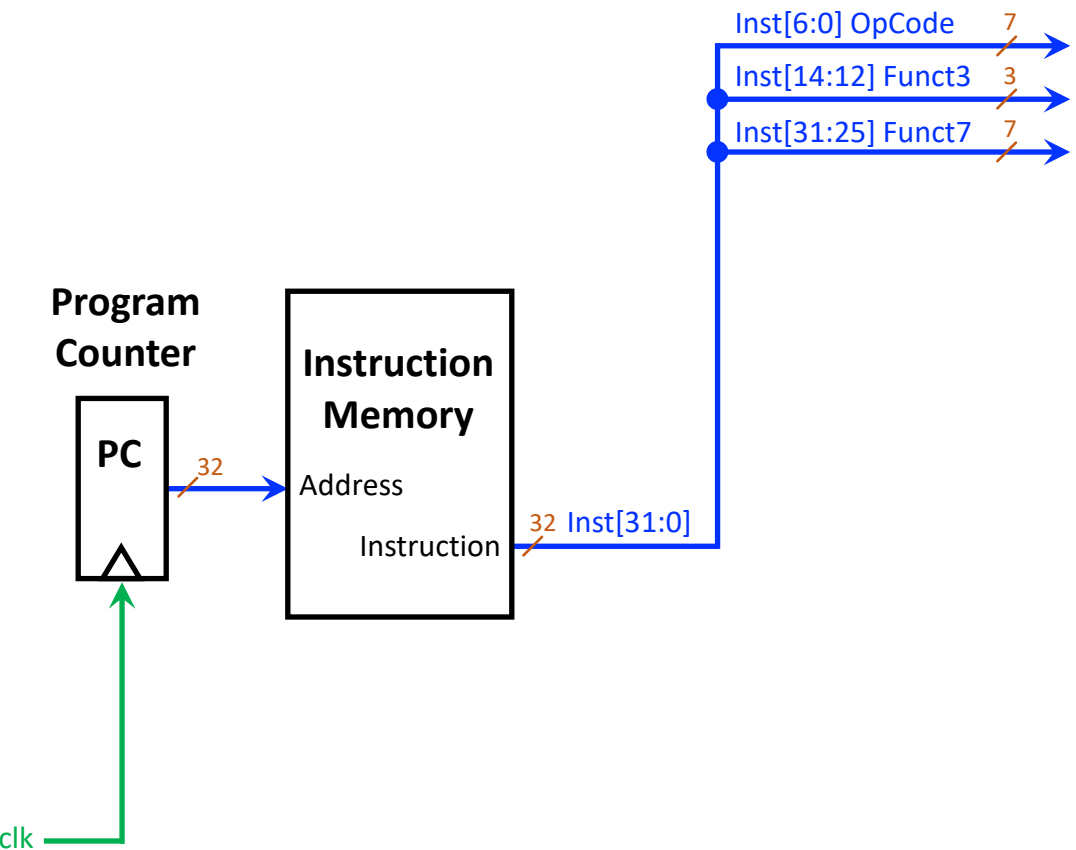
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R			Funct7				rs2				rs1			Funct3			rd				OpCode	
I						Imm[11:0]					rs1			Funct3			rd				OpCode	
S						Imm[11:5]			rs2		rs1			Funct3			Imm[4:0]				OpCode	
B						Imm[12,10:5]			rs2		rs1			Funct3			Imm[4:1,11]				OpCode	
U									Imm[31:12]								rd				OpCode	
J									Imm[20,10:1,11,19:12]								rd				OpCode	



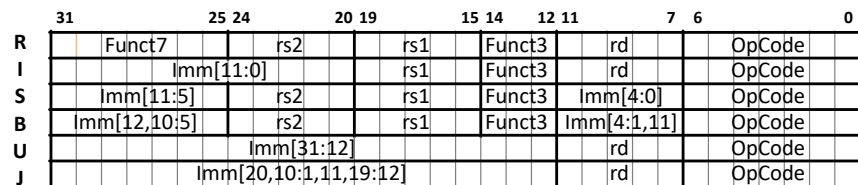
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R		Funct7			rs2			rs1		Funct3			rd			OpCode
I		Imm[11:0]					rs1		Funct3		rd					OpCode
S		Imm[11:5]				rs2		rs1		Funct3	Imm[4:0]					OpCode
B		Imm[12,10:5]				rs2		rs1		Funct3	Imm[4:1,11]					OpCode
U		Imm[31:12]										rd				OpCode
J		Imm[20,10:1,11,19:12]										rd				OpCode



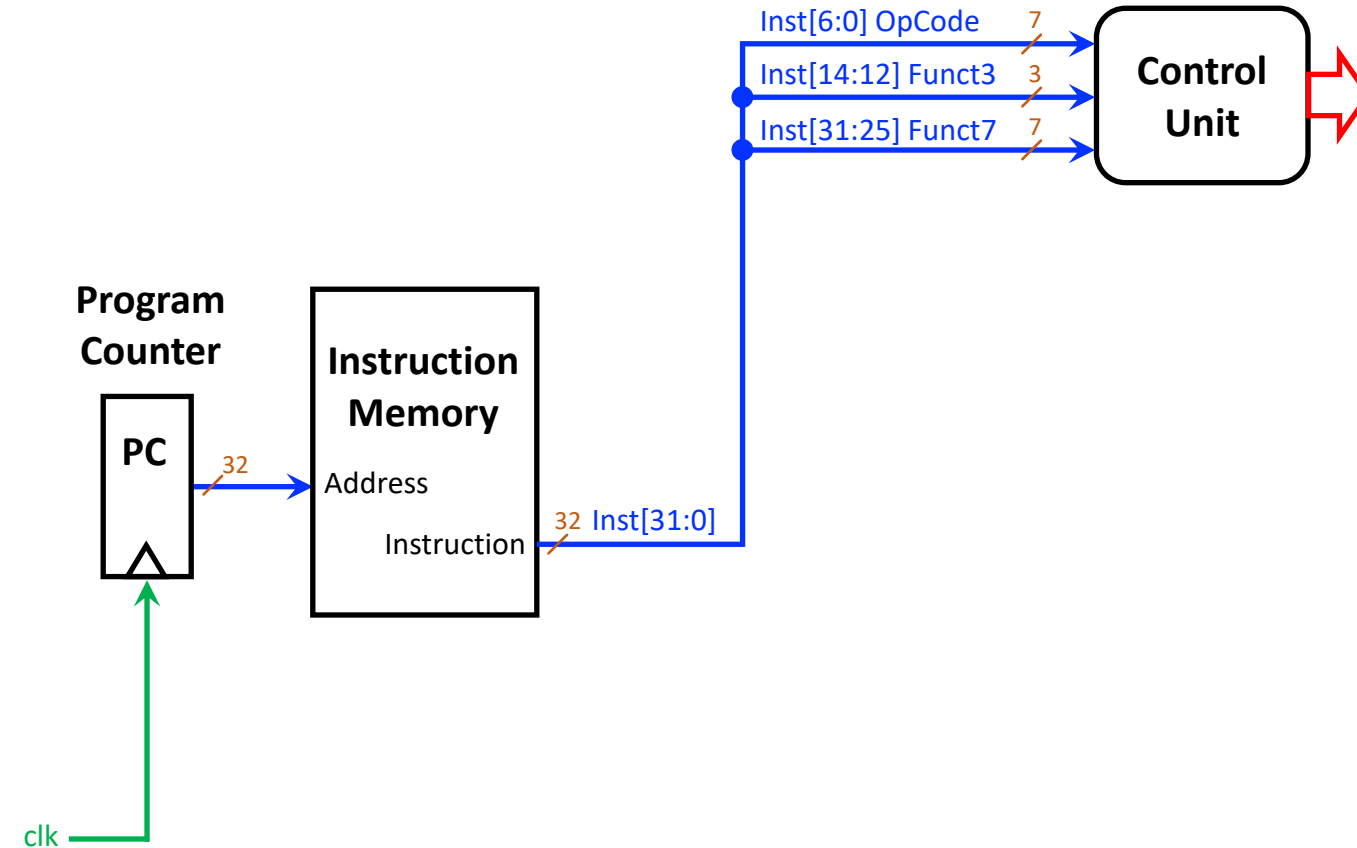
	31	25	24	20	19	15	14	12	11	7	6	0				
R		Funct7			rs2			rs1		Funct3			rd			OpCode
I		Imm[11:0]					rs1		Funct3		rd					OpCode
S		Imm[11:5]				rs2		rs1		Funct3	Imm[4:0]					OpCode
B		Imm[12,10:5]				rs2		rs1		Funct3	Imm[4:1,11]					OpCode
U		Imm[31:12]										rd				OpCode
J		Imm[20,10:1,11,19:12]										rd				OpCode



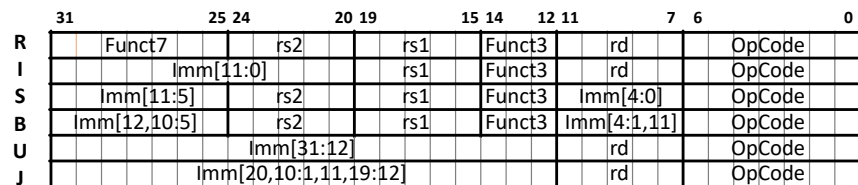
	31	25	24	20	19	15	14	12	11	7	6	0
R												
I												
S												
B												
U												
J												

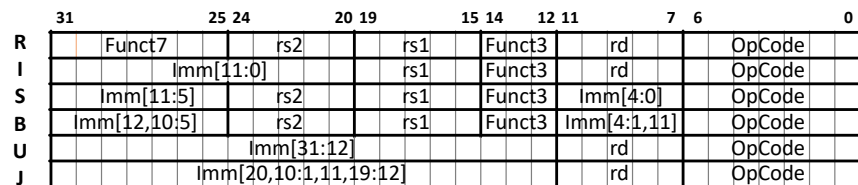


Procesador RISC-V Monociclo

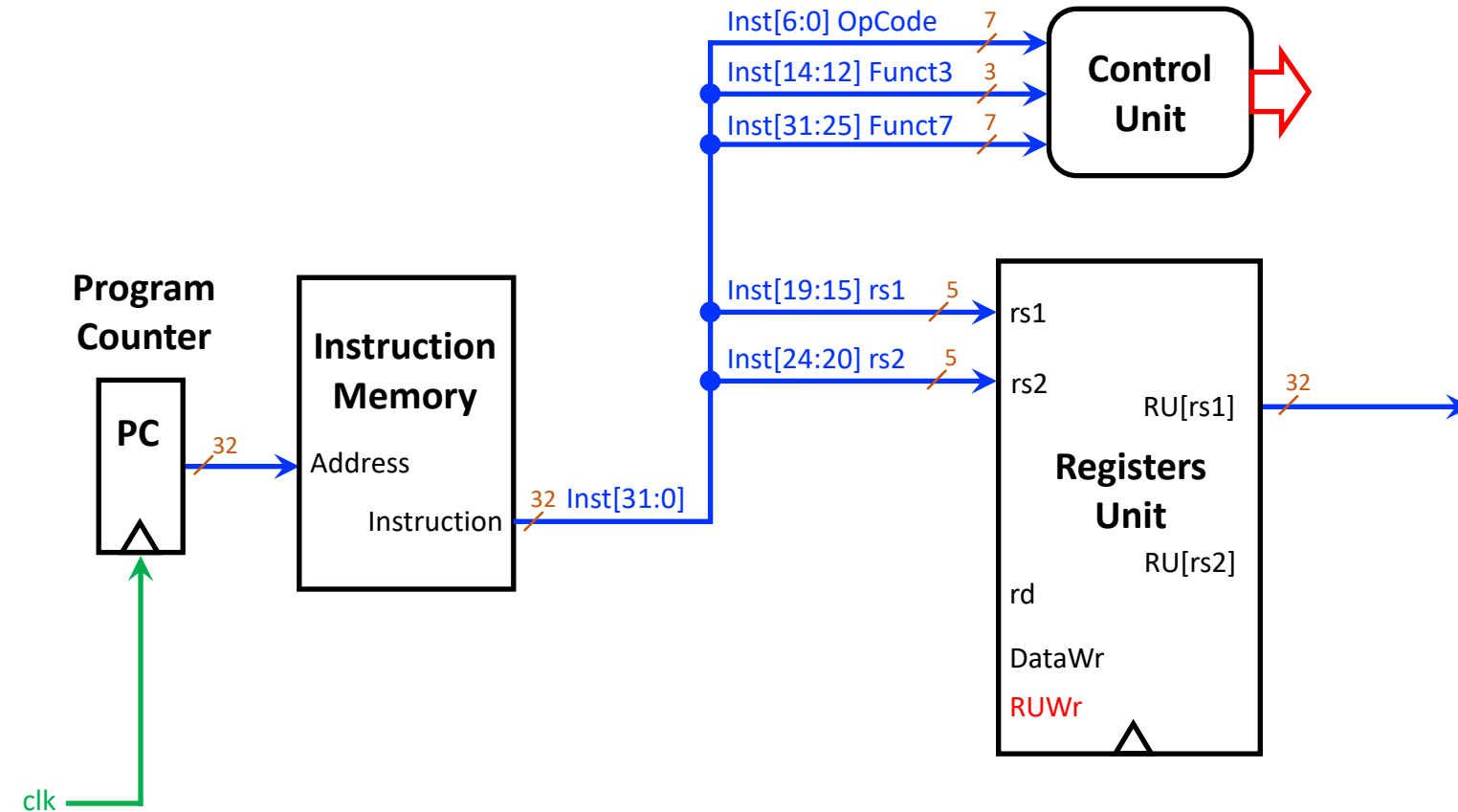


	31	25	24	20	19	15	14	12	11	7	6	0
R												
I												
S												
B												
U												
J												

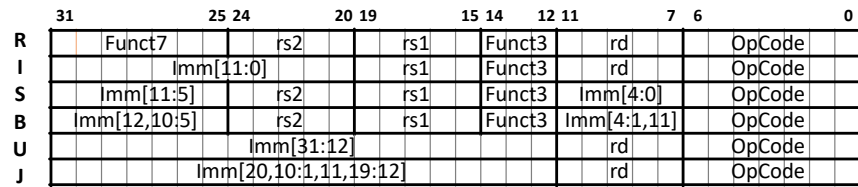




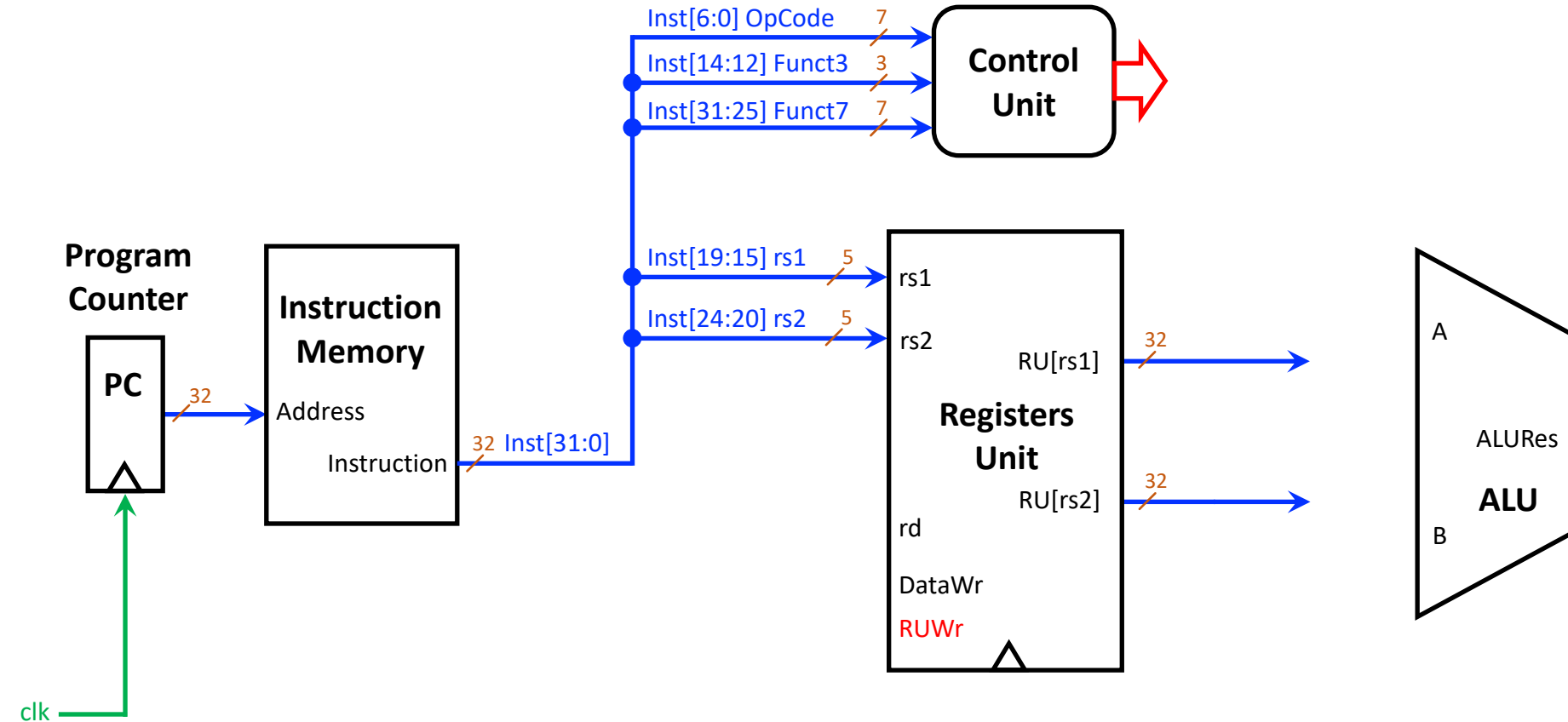
Procesador RISC-V Monociclo



	31	25	24	20	19	15	14	12	11	7	6	0
R												
I												
S												
B												
U												
J												

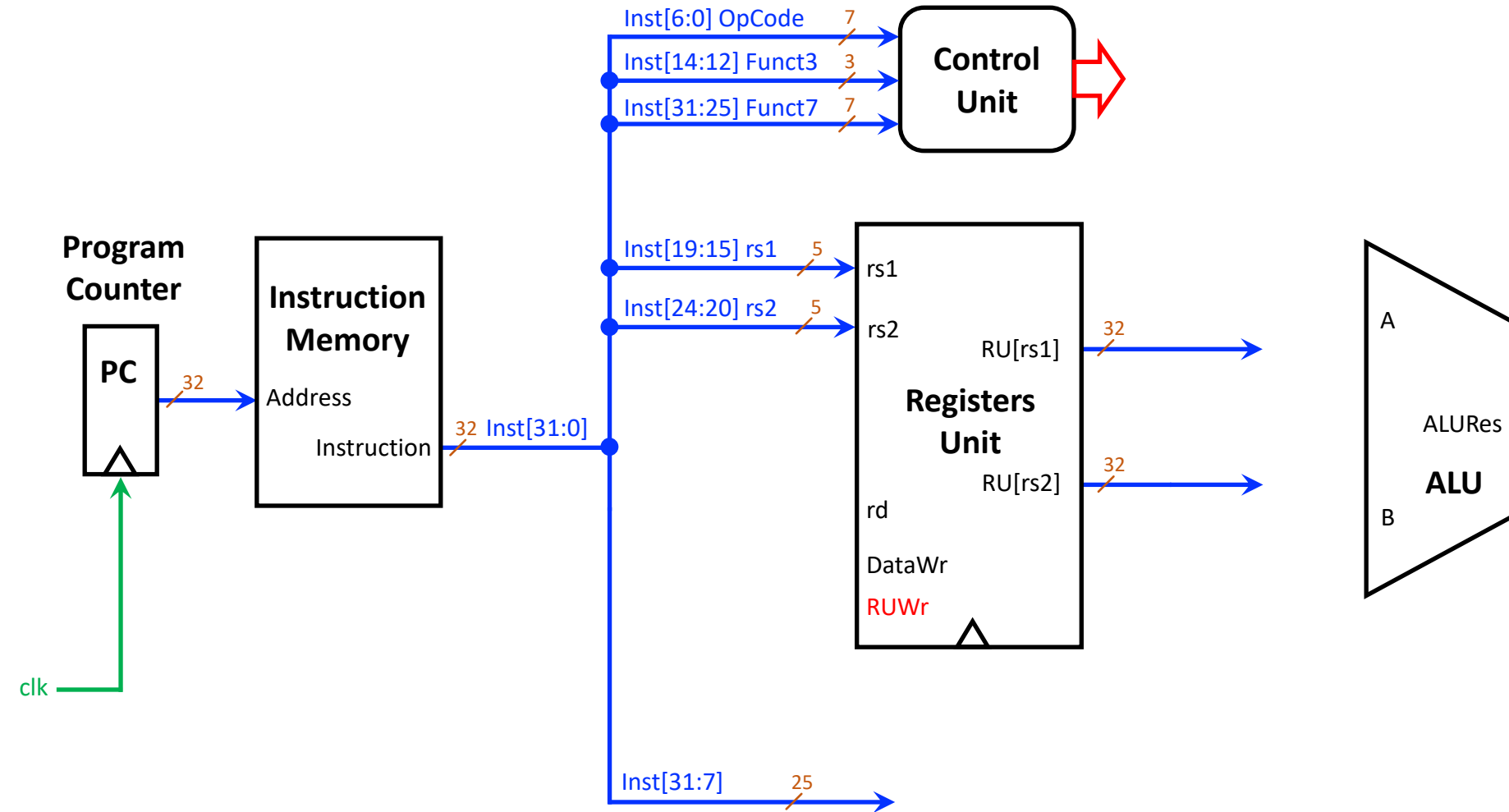


Procesador RISC-V Monociclo



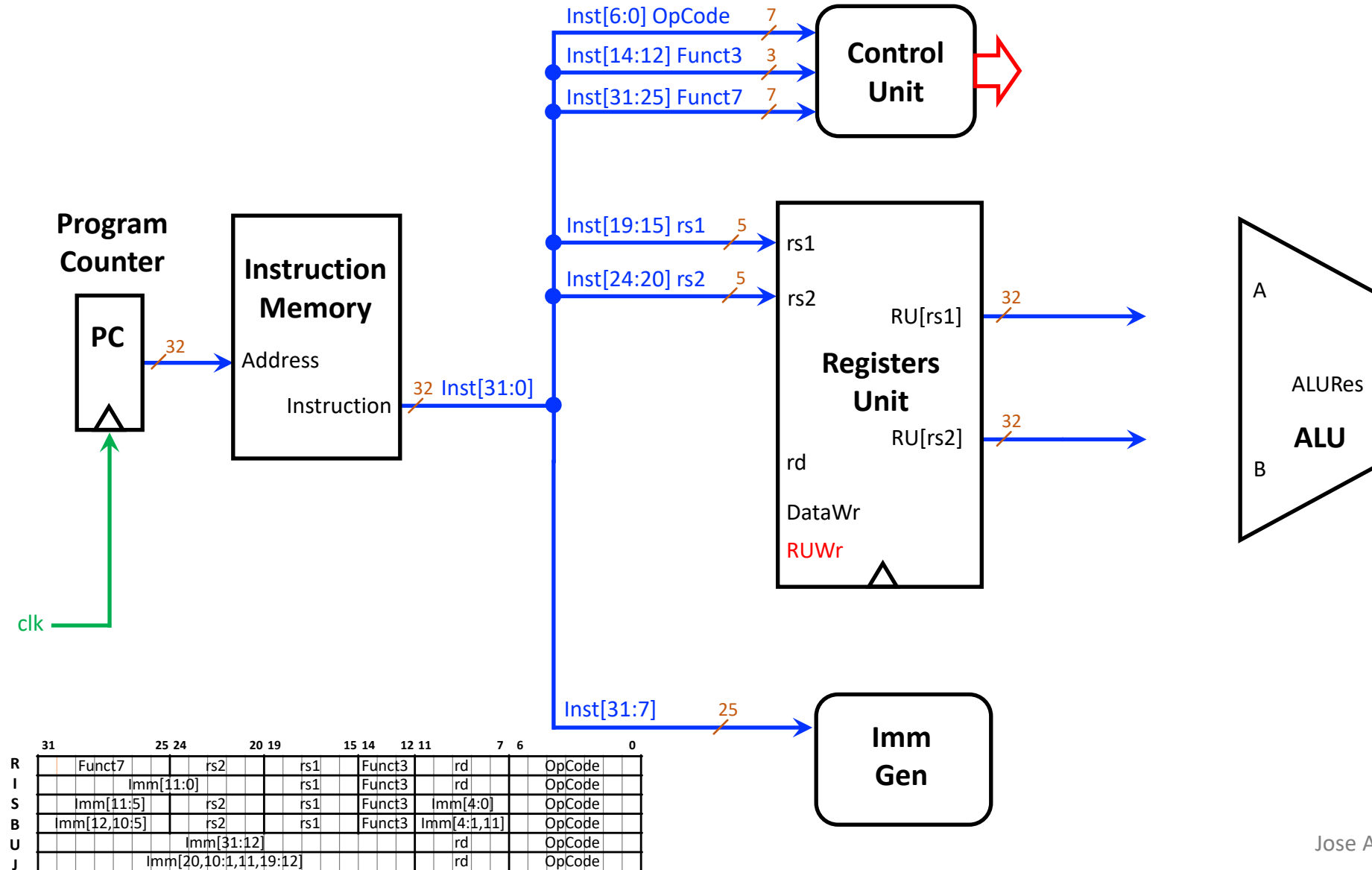
	31	25	24	20	19	15	14	12	11	7	6	0
R												
I												
S												
B												
U												
J												

Procesador RISC-V Monociclo

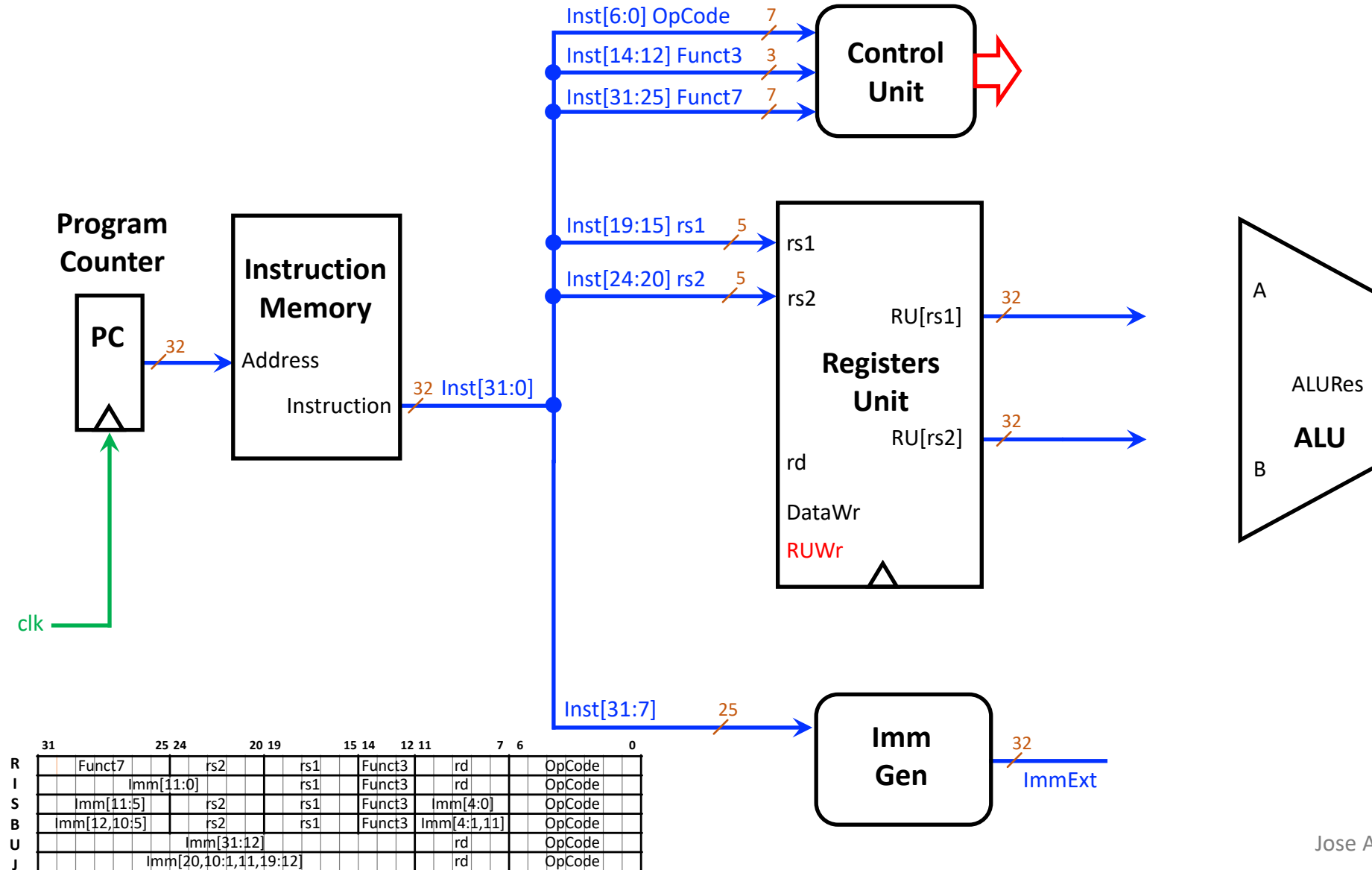


	31	25	24	20	19	15	14	12	11	7	6	0
R												
I												
S												
B												
U												
J												

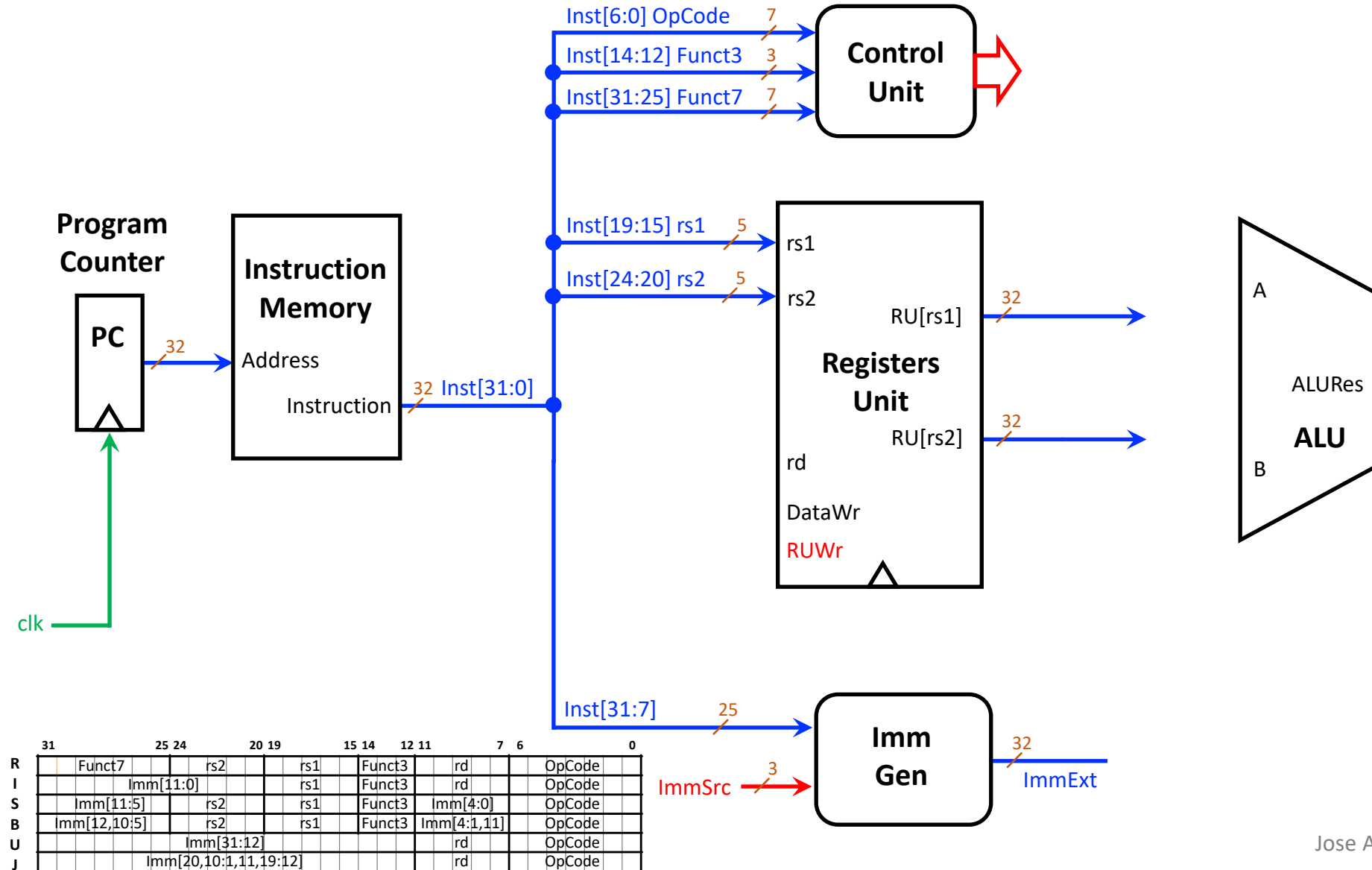
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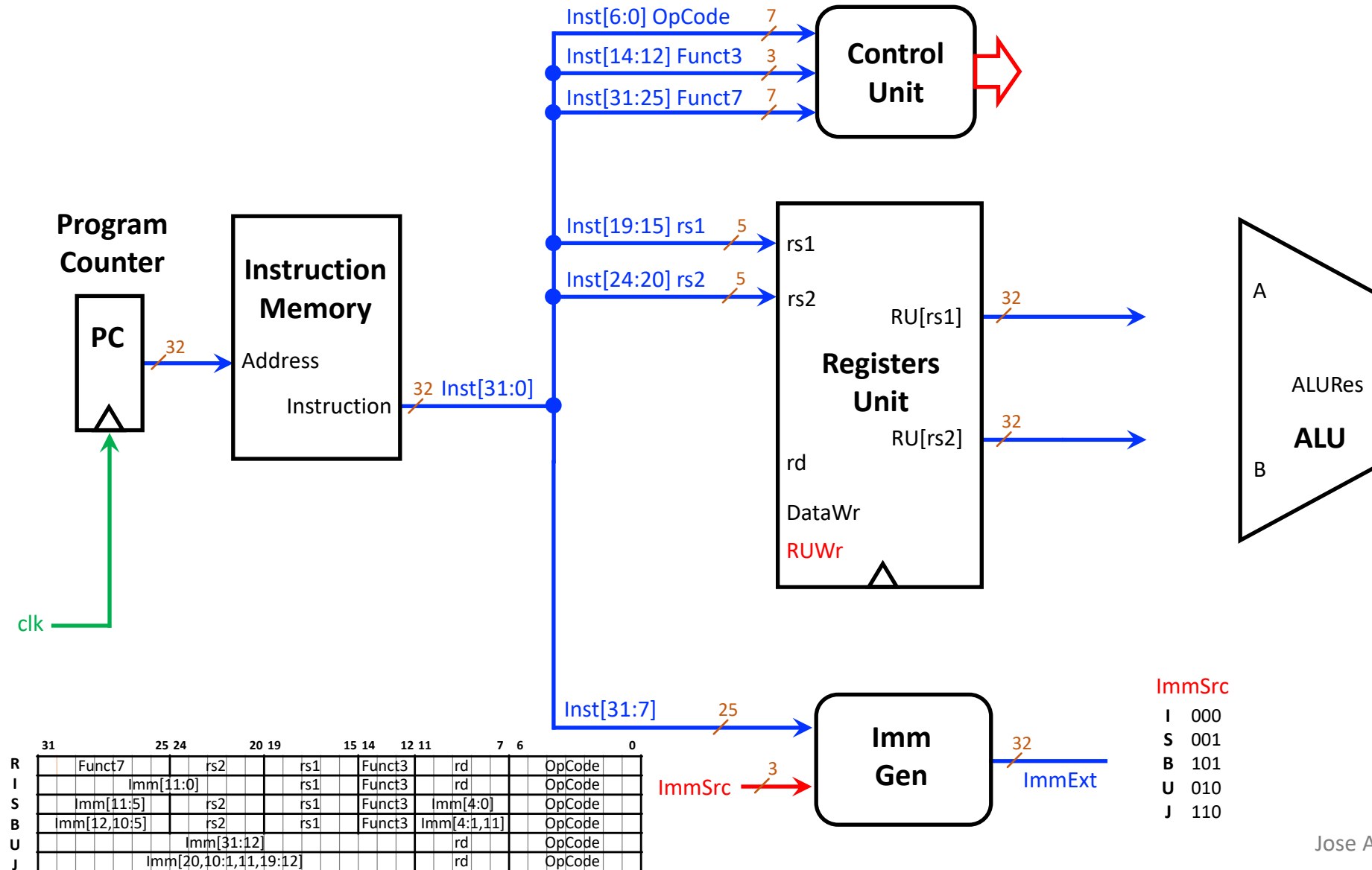
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Procesador RISC-V Monociclo



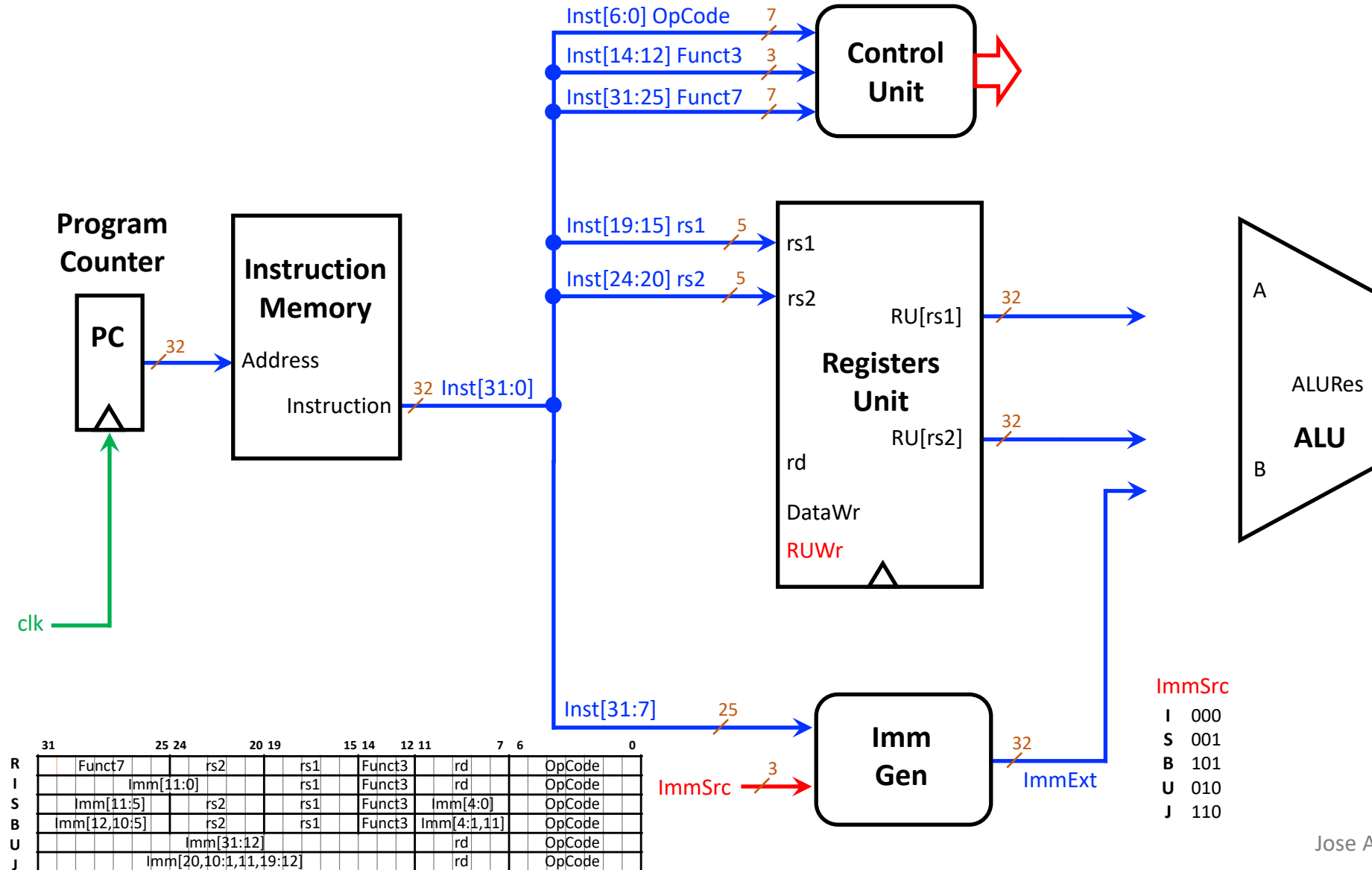
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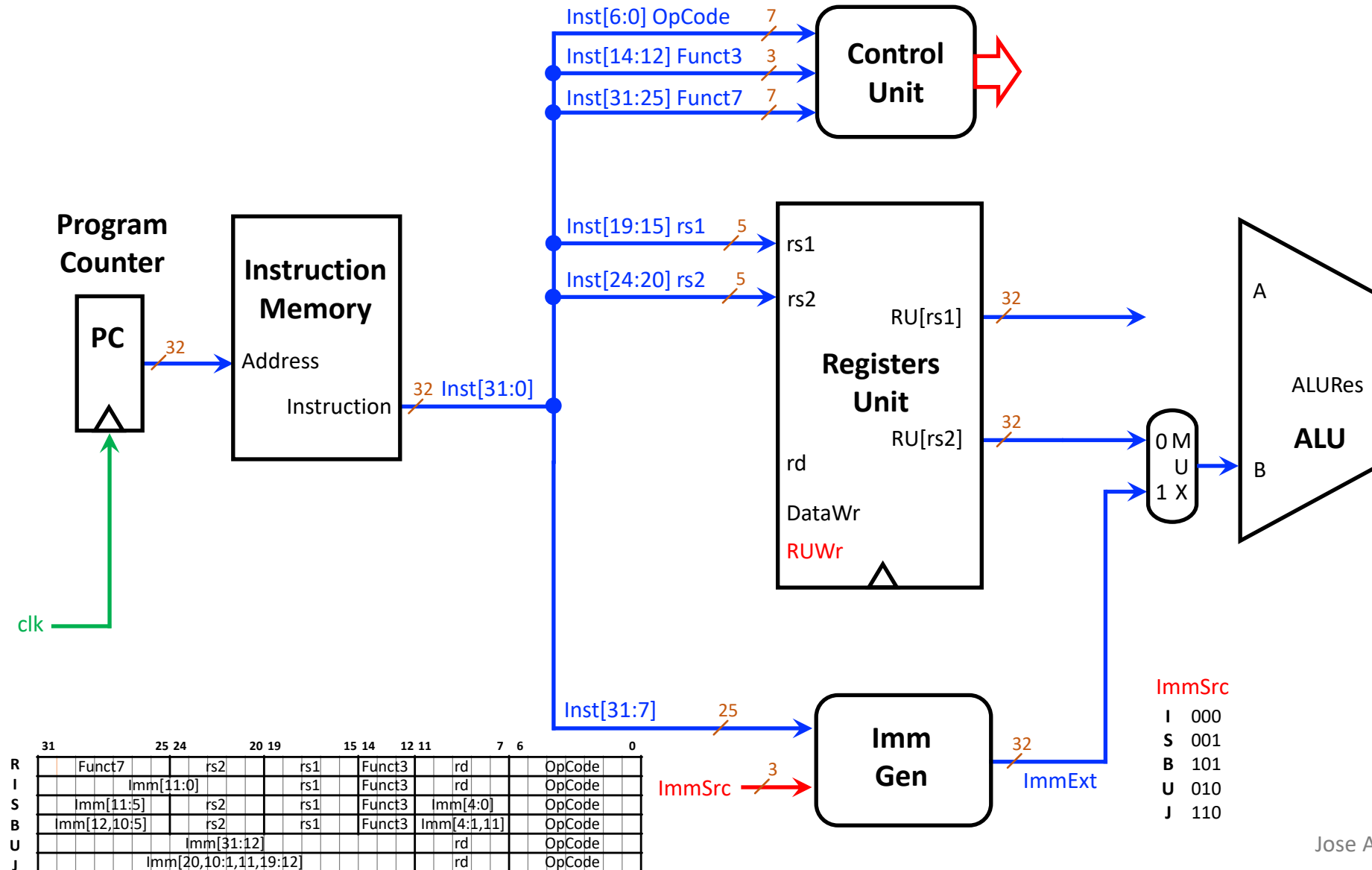
ImmSrc

I 000
S 001
B 101
U 010
J 110

Procesador RISC-V Monociclo



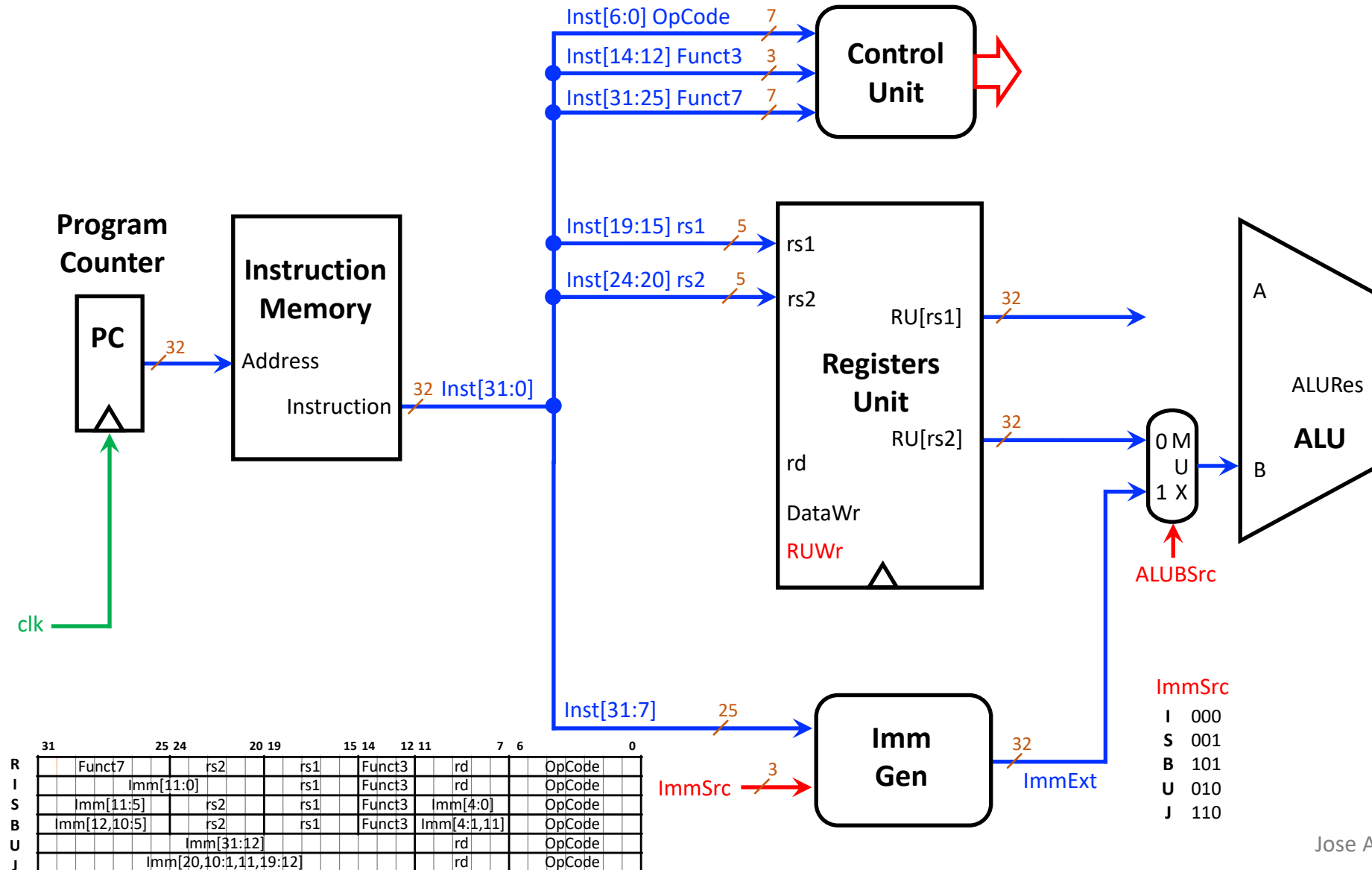
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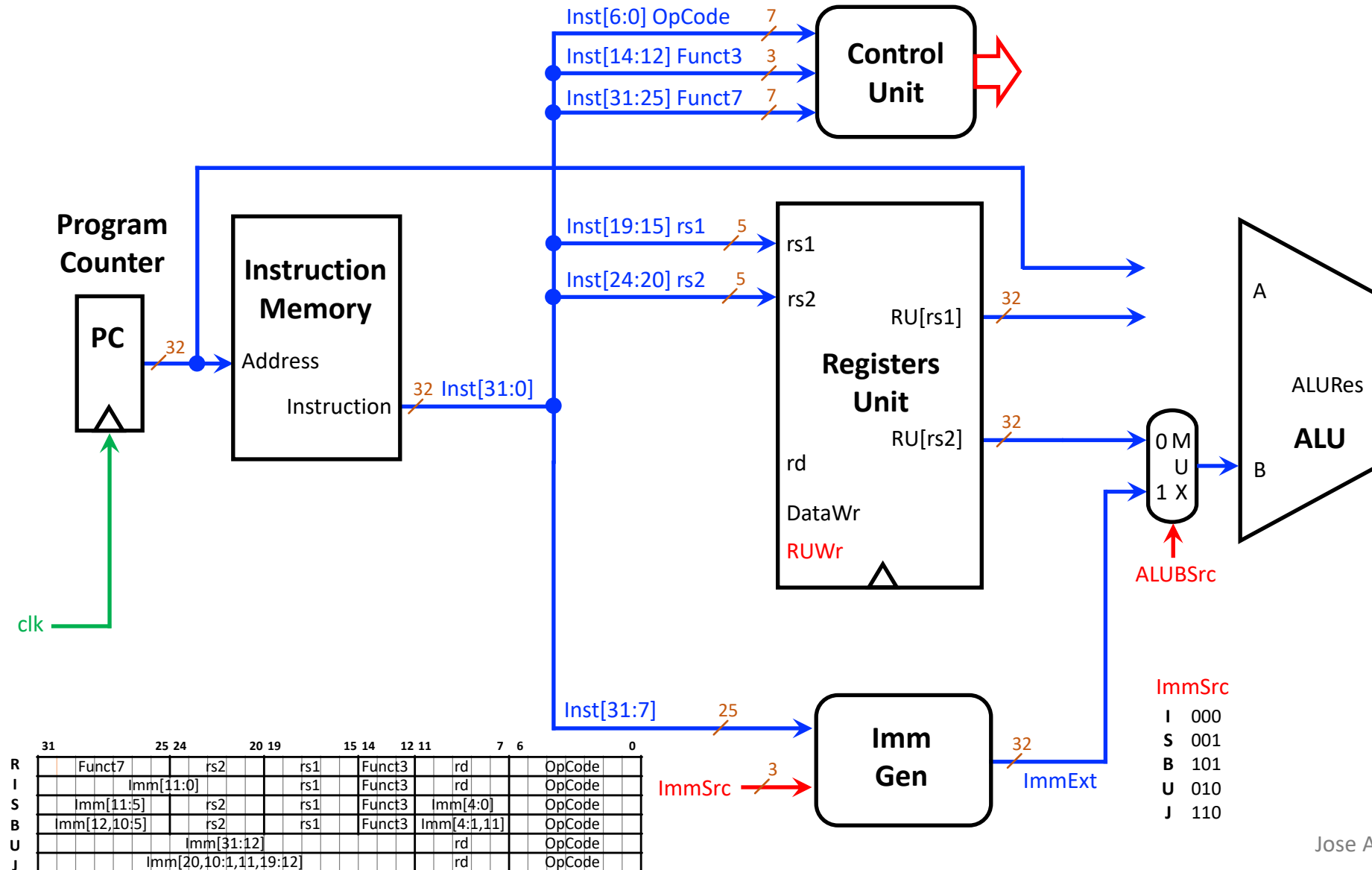
ImmSrc

I 000
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B 101
U 010
J 110

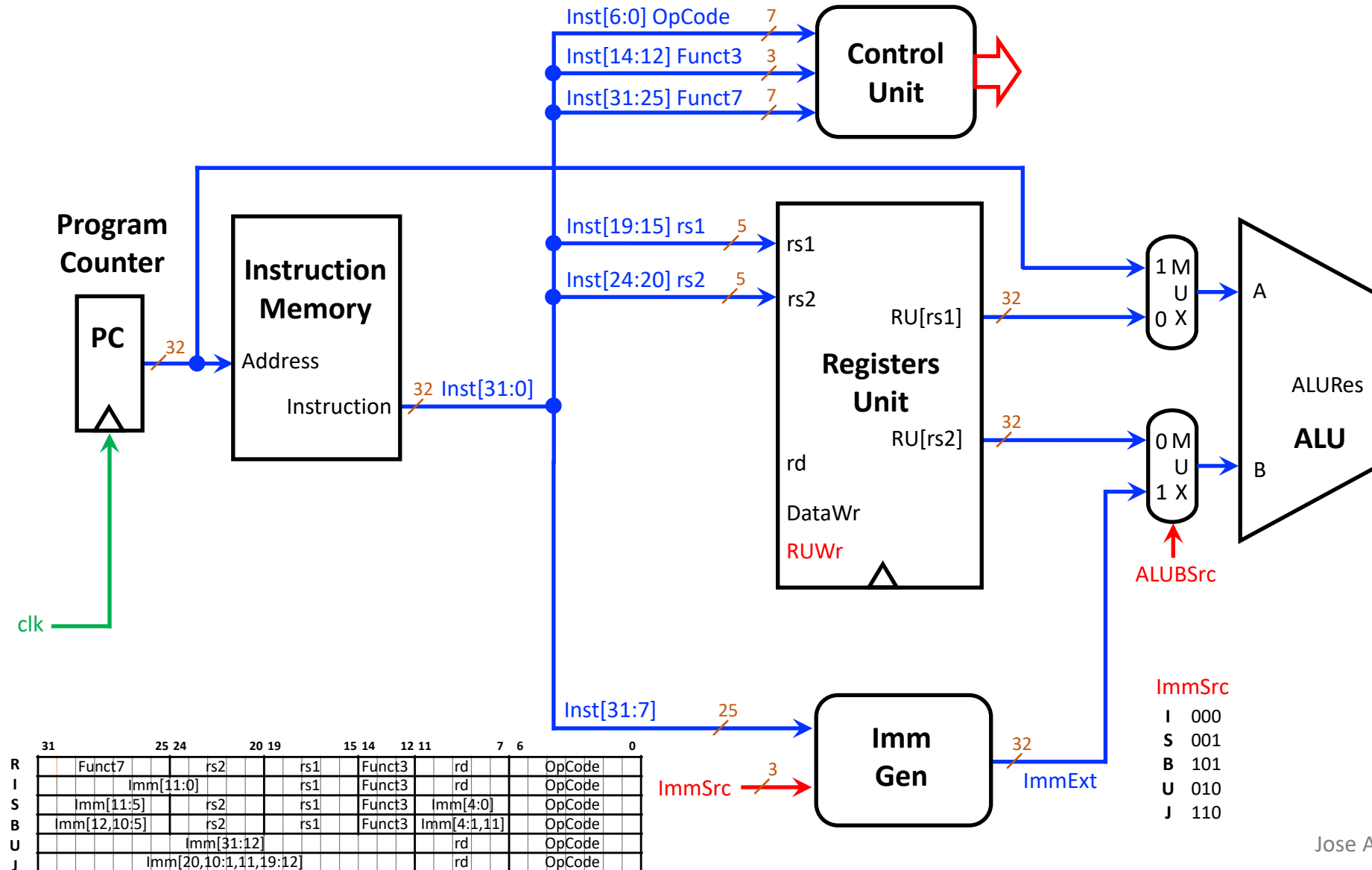
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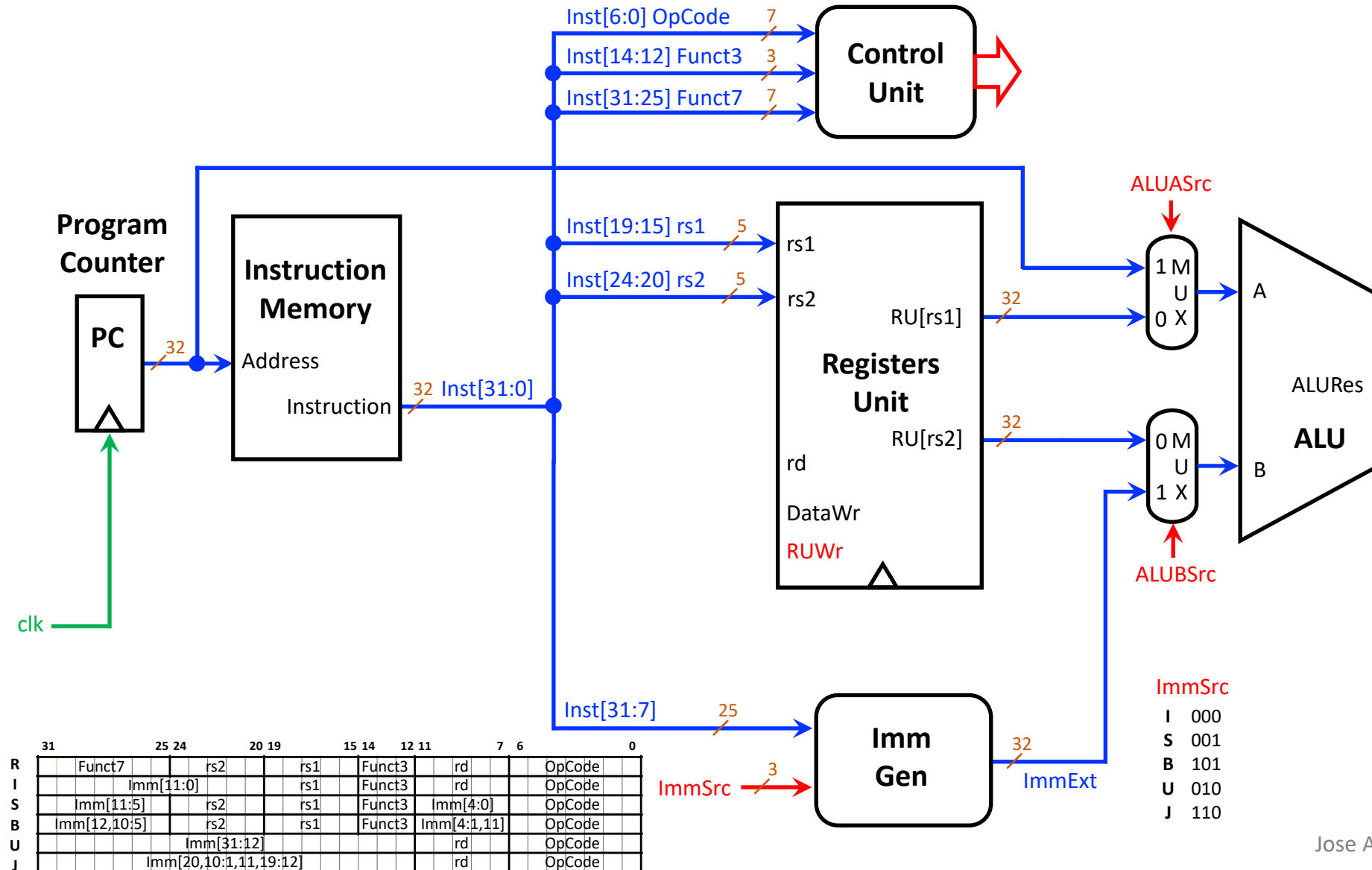
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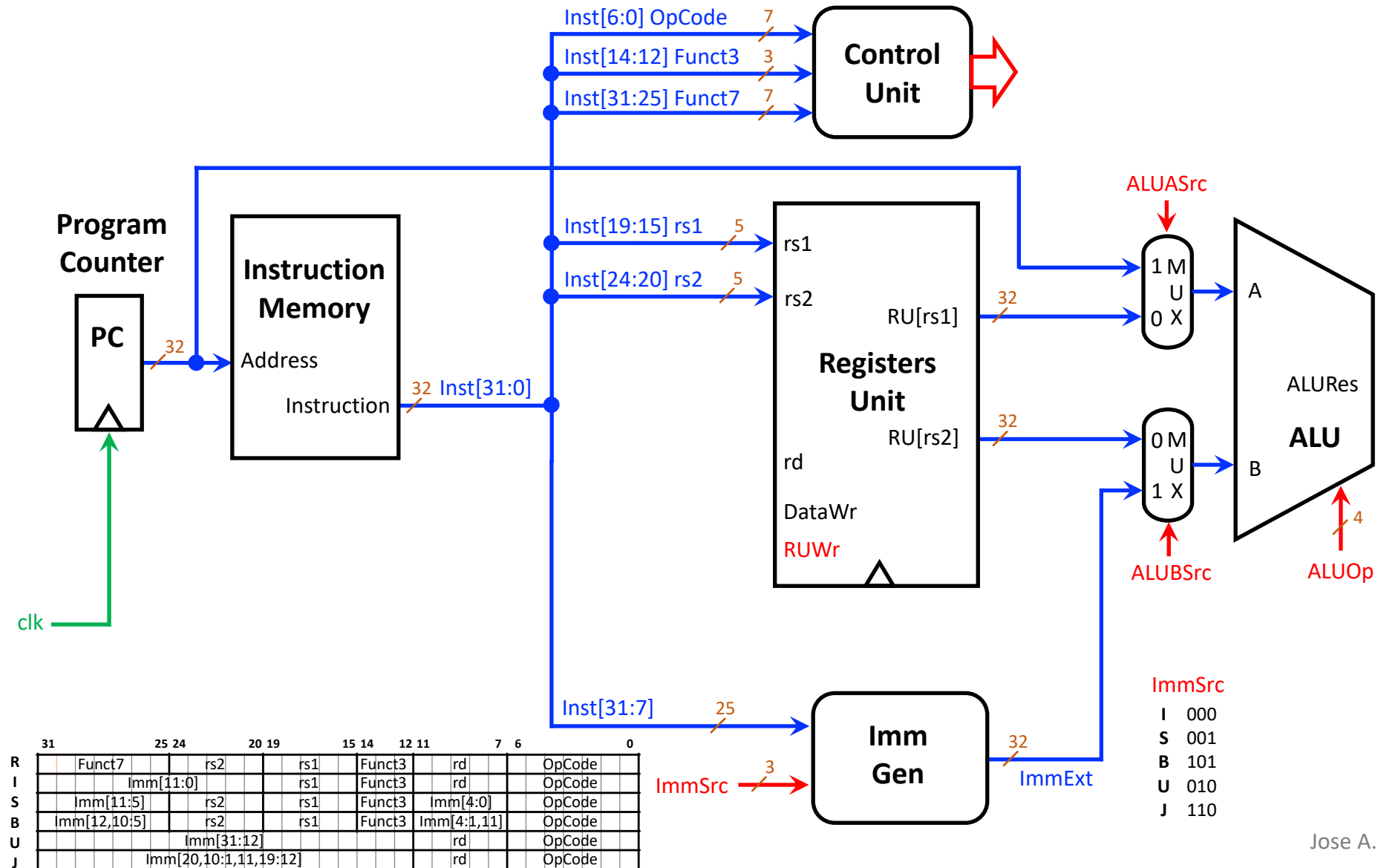
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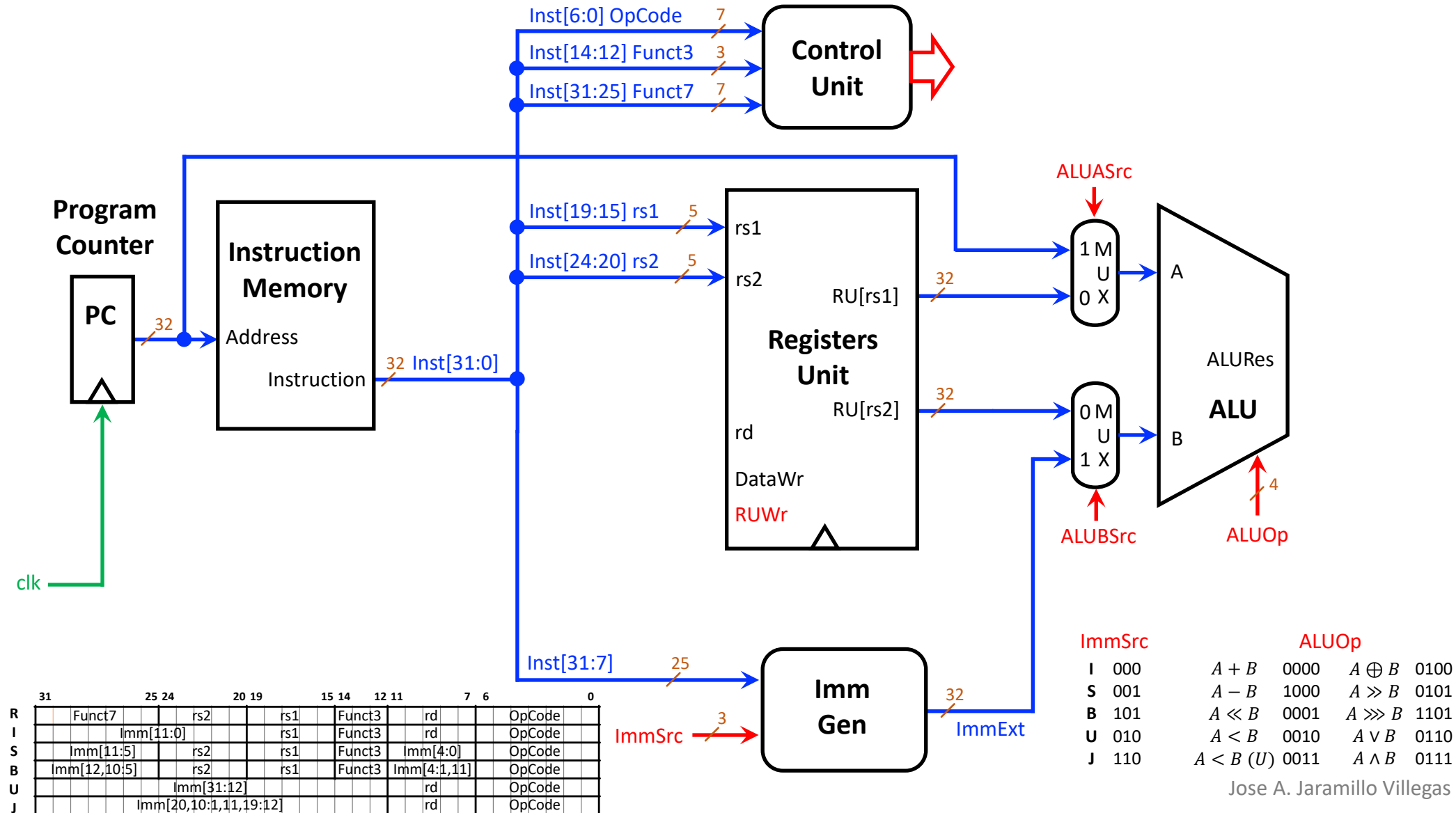
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Procesador RISC-V Monociclo

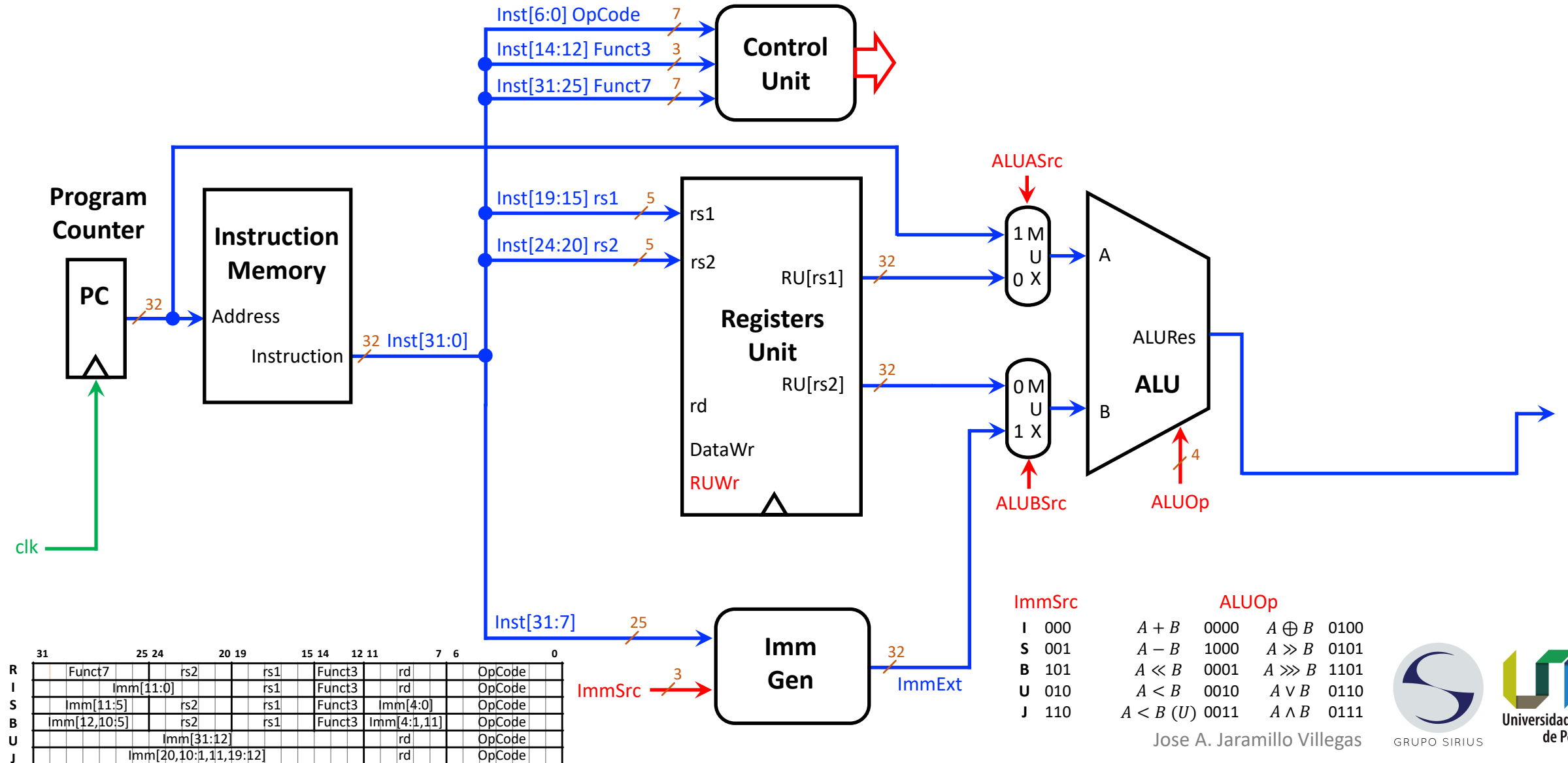


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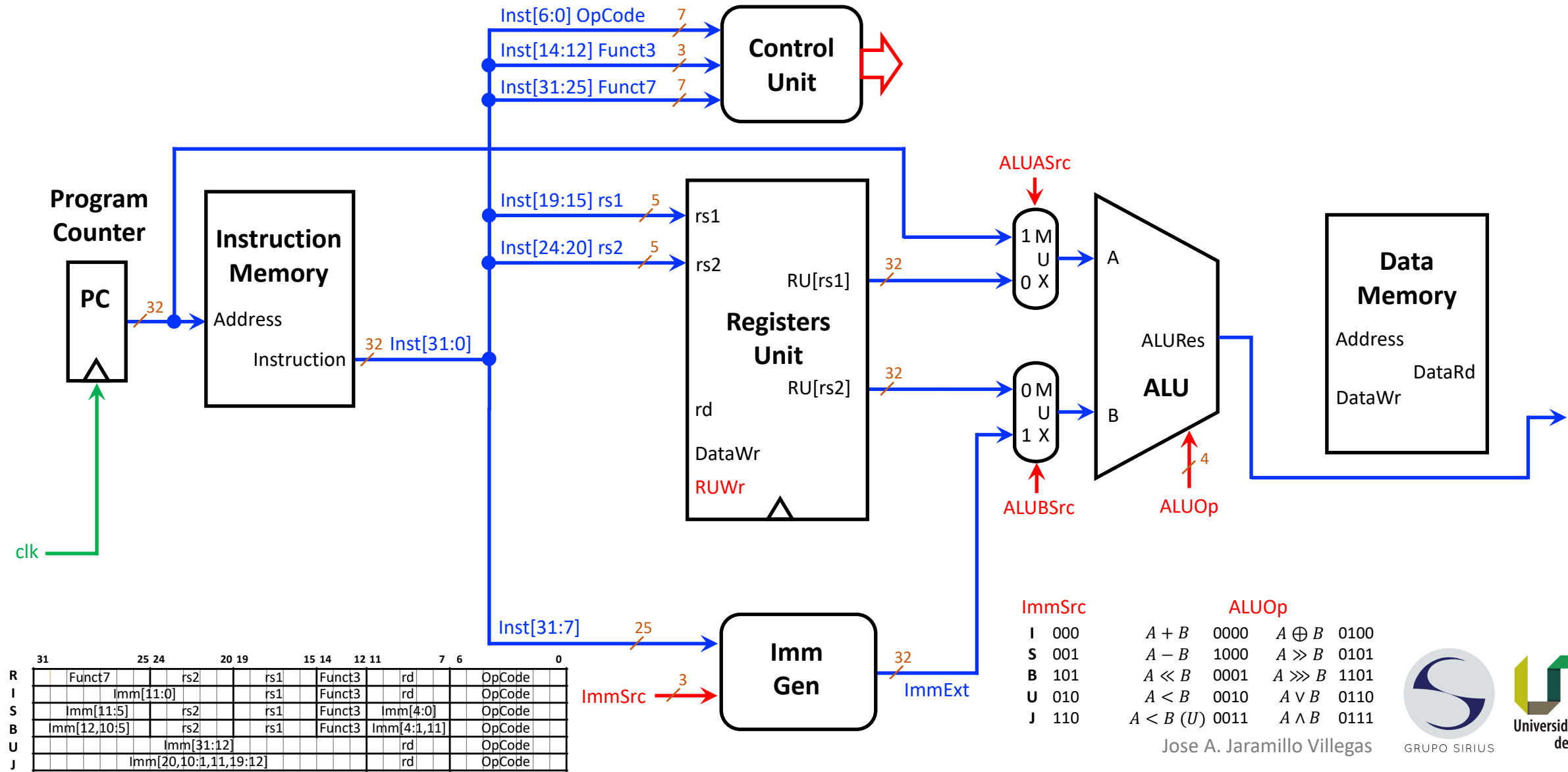


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Procesador RISC-V Monociclo



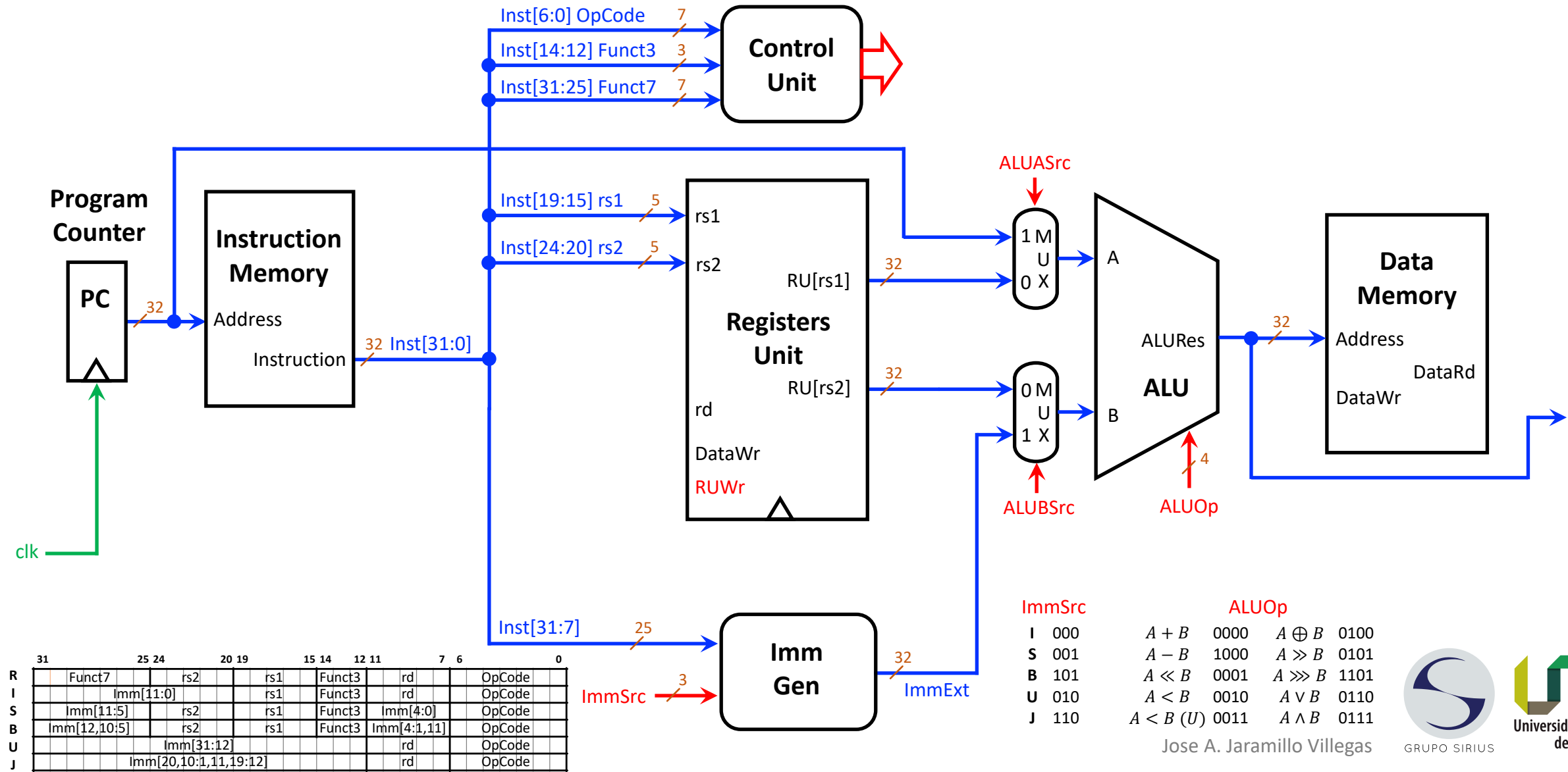
Procesador
RISC-V
Monociclo



ImmSrc		ALUOp	
I	000	$A + B$	0000
S	001	$A - B$	1000
B	101	$A \ll B$	0001
U	010	$A < B$	0010
J	110	$A < B (U)$	0011
		$A \oplus B$	0100
		$A \gg B$	0101
		$A \ggg B$	1101
		$A \vee B$	0110
		$A \wedge B$	0111

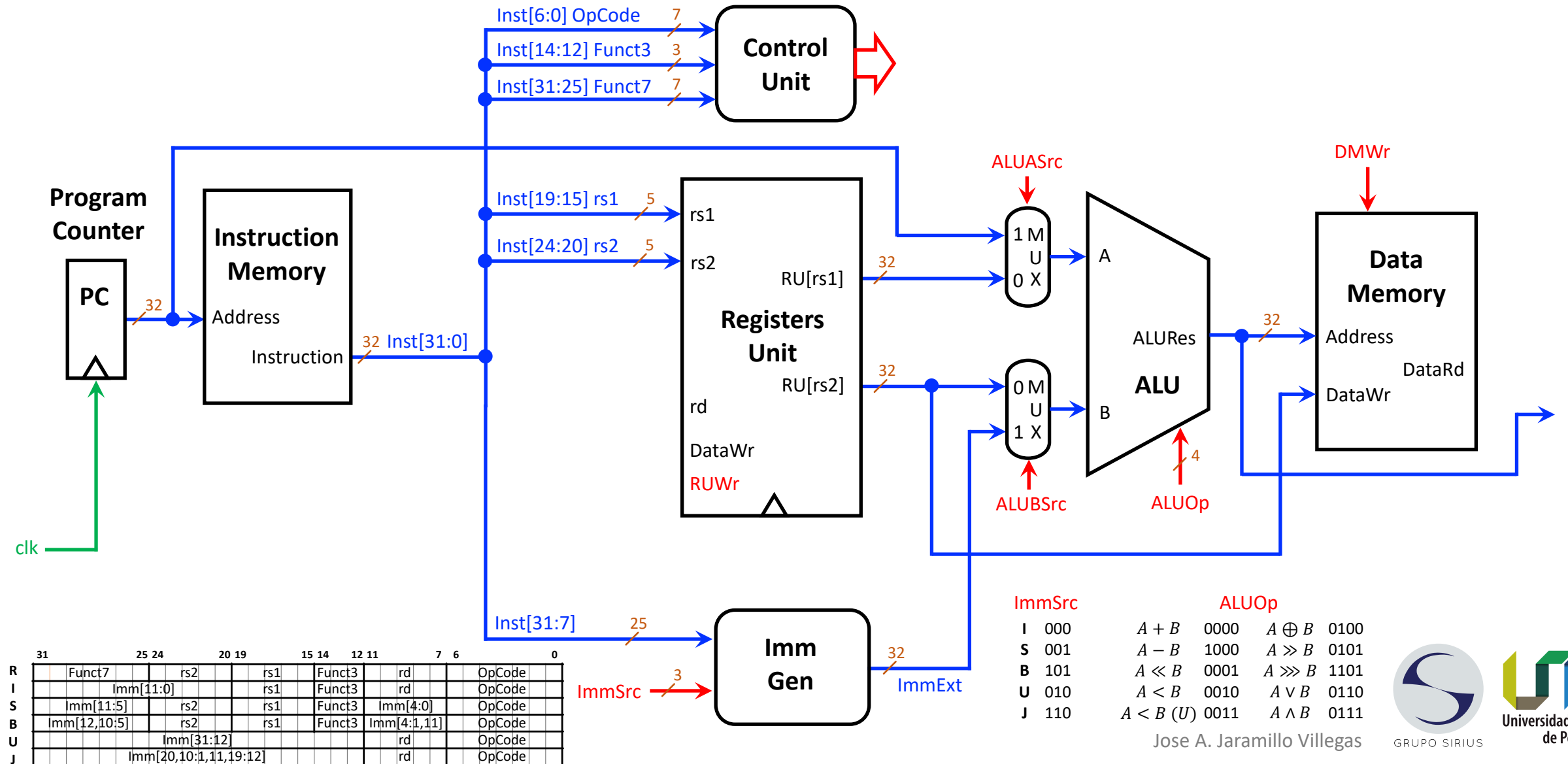
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Procesador
RISC-V
Monociclo

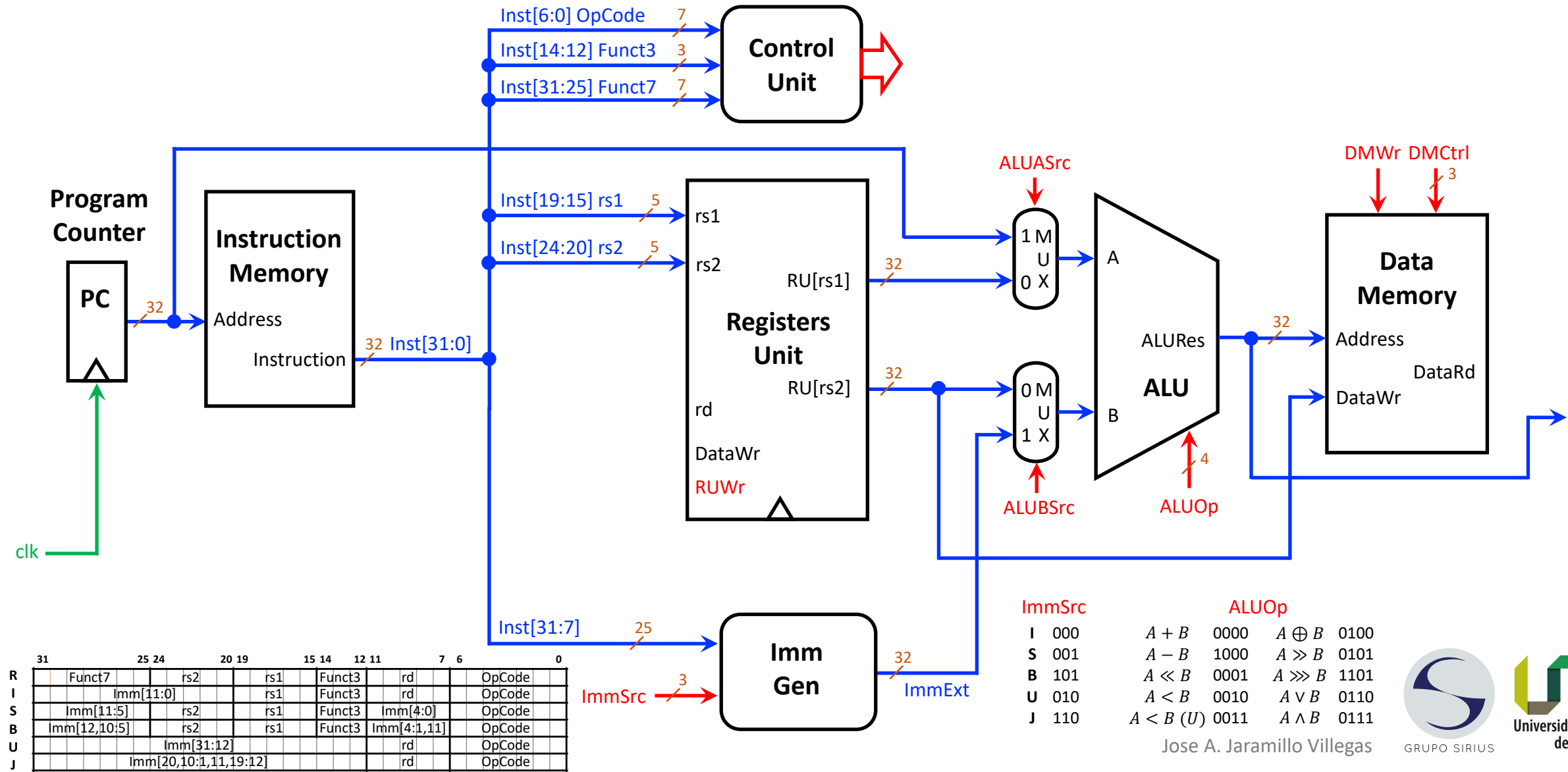




Procesador RISC-V Monociclo



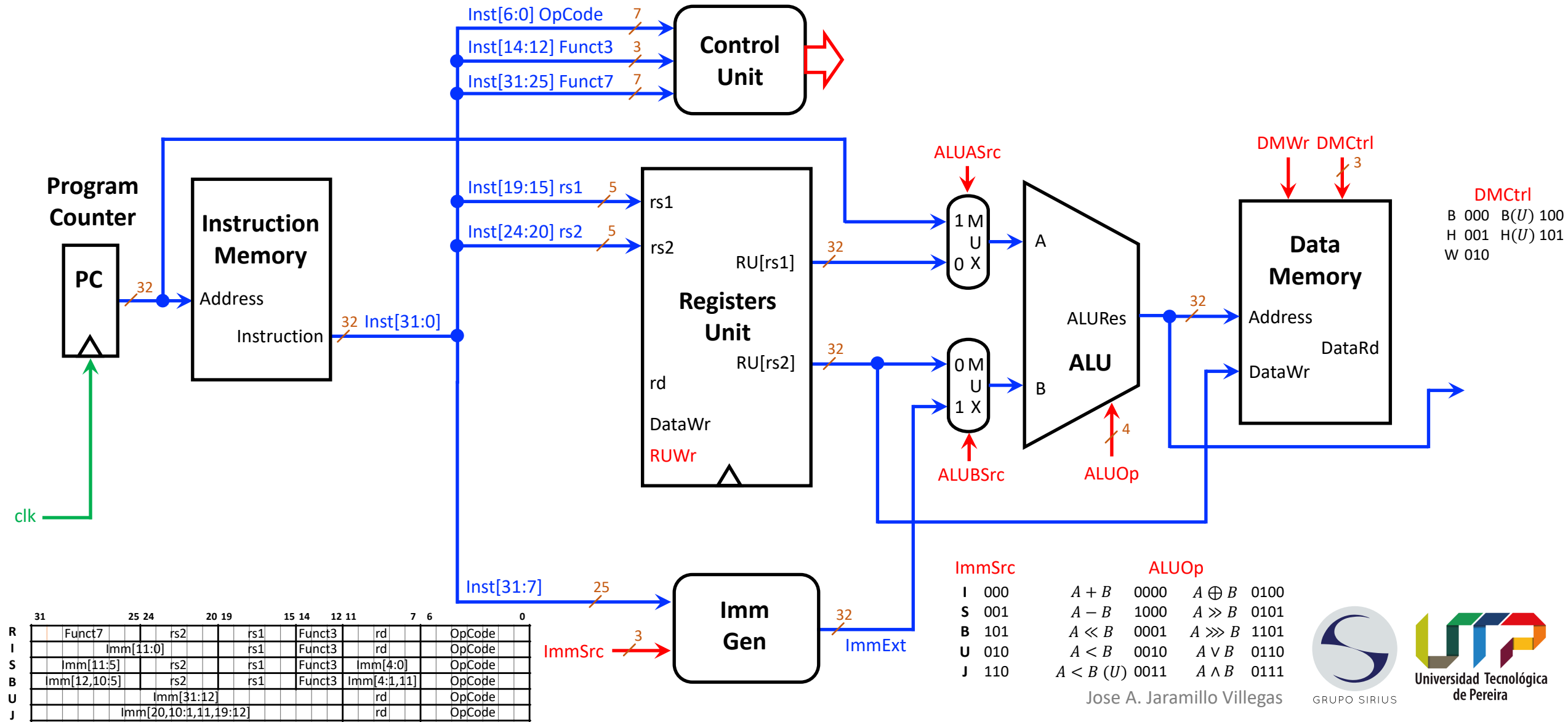
Procesador
RISC-V
Monociclo



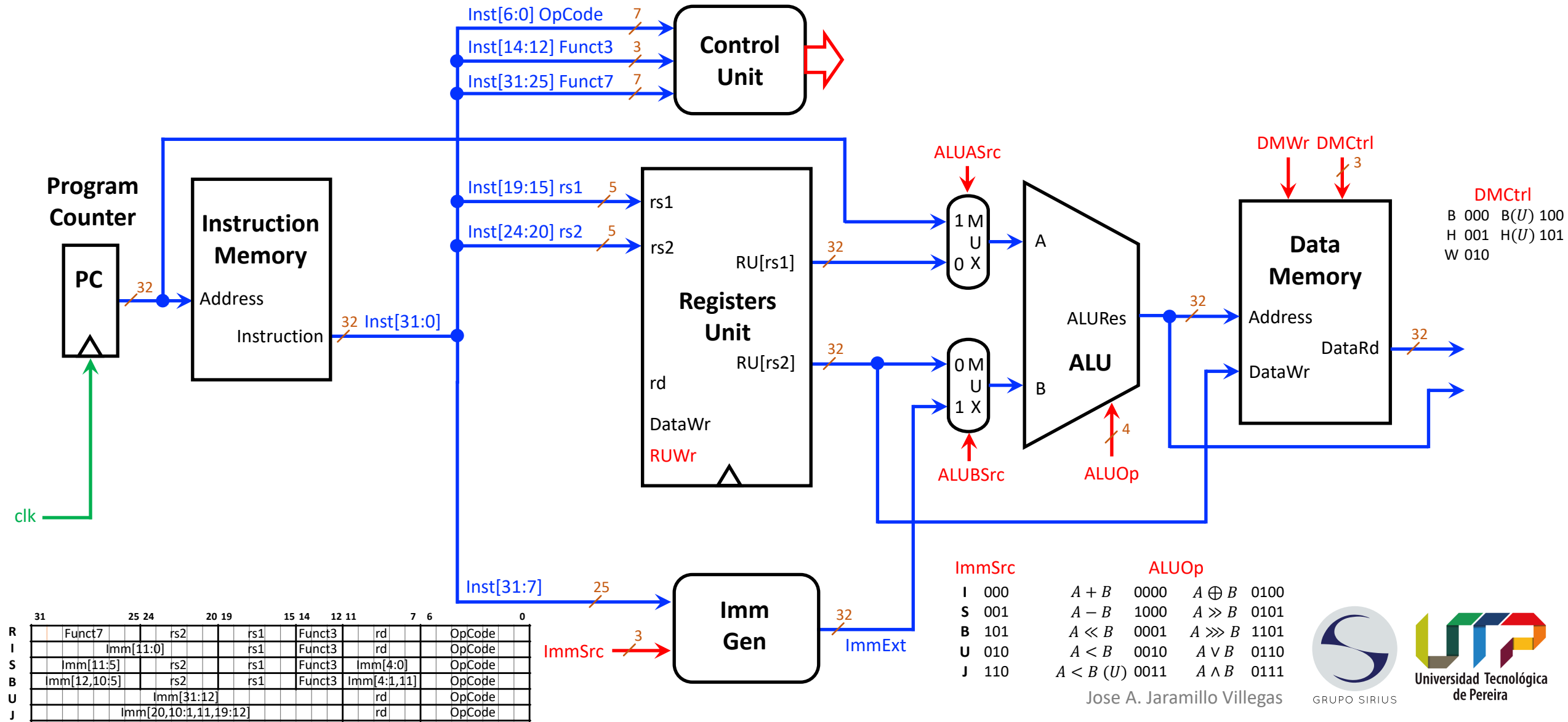
ImmSrc		ALUOp	
I	000	$A + B$	0000
S	001	$A - B$	1000
B	101	$A \ll B$	0001
U	010	$A < B$	0010
J	110	$A < B (U)$	0011
		$A \oplus B$	0100
		$A \gg B$	0101
		$A \ggg B$	1101
		$A \vee B$	0110
		$A \wedge B$	0111

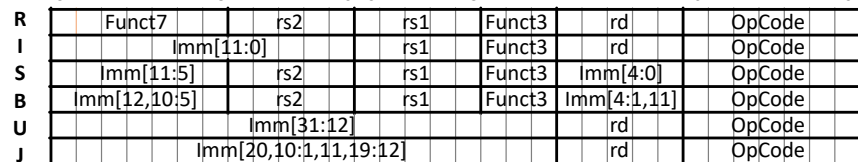
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Procesador
RISC-V
Monociclo

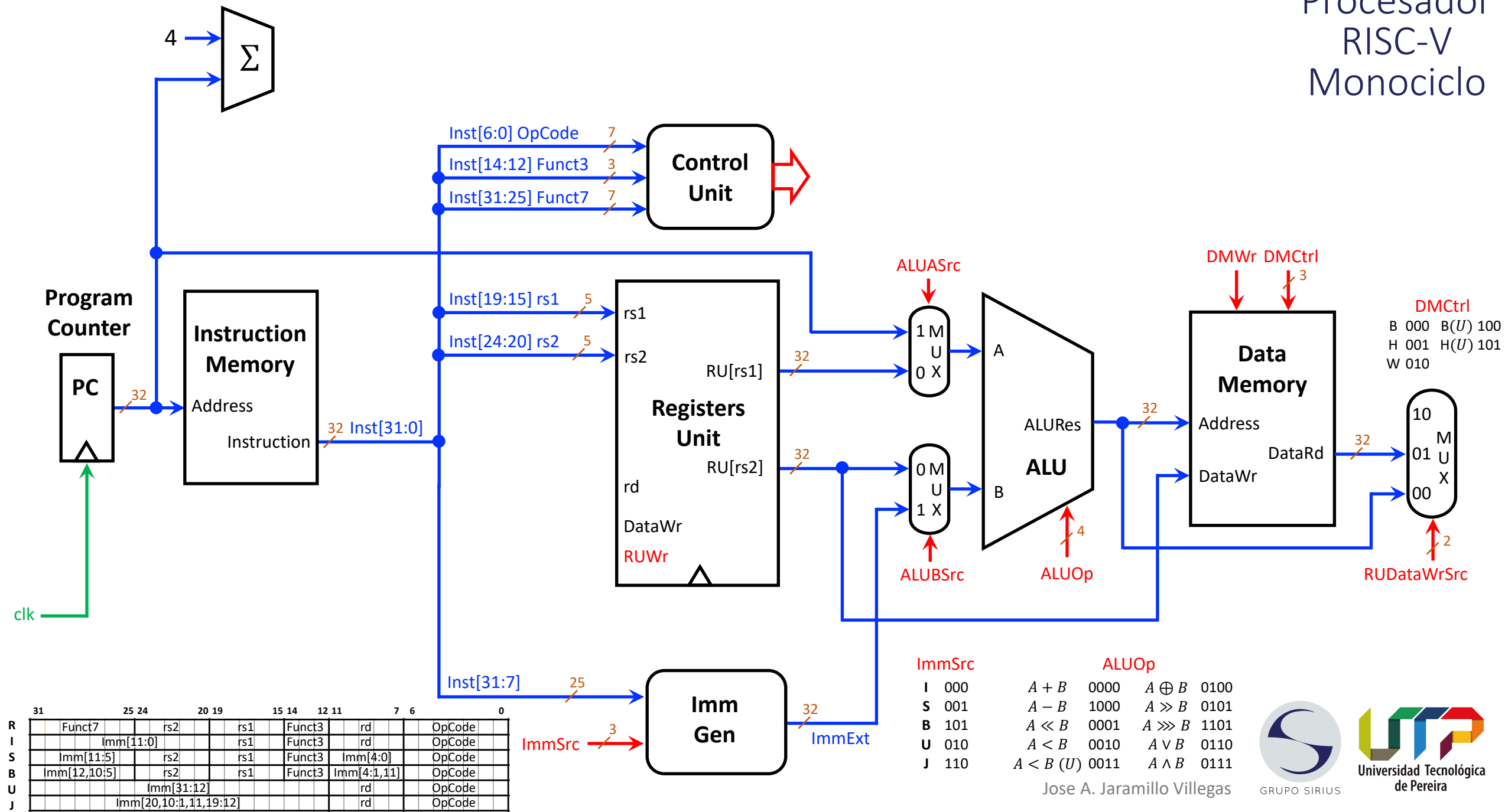


Procesador RISC-V Monociclo

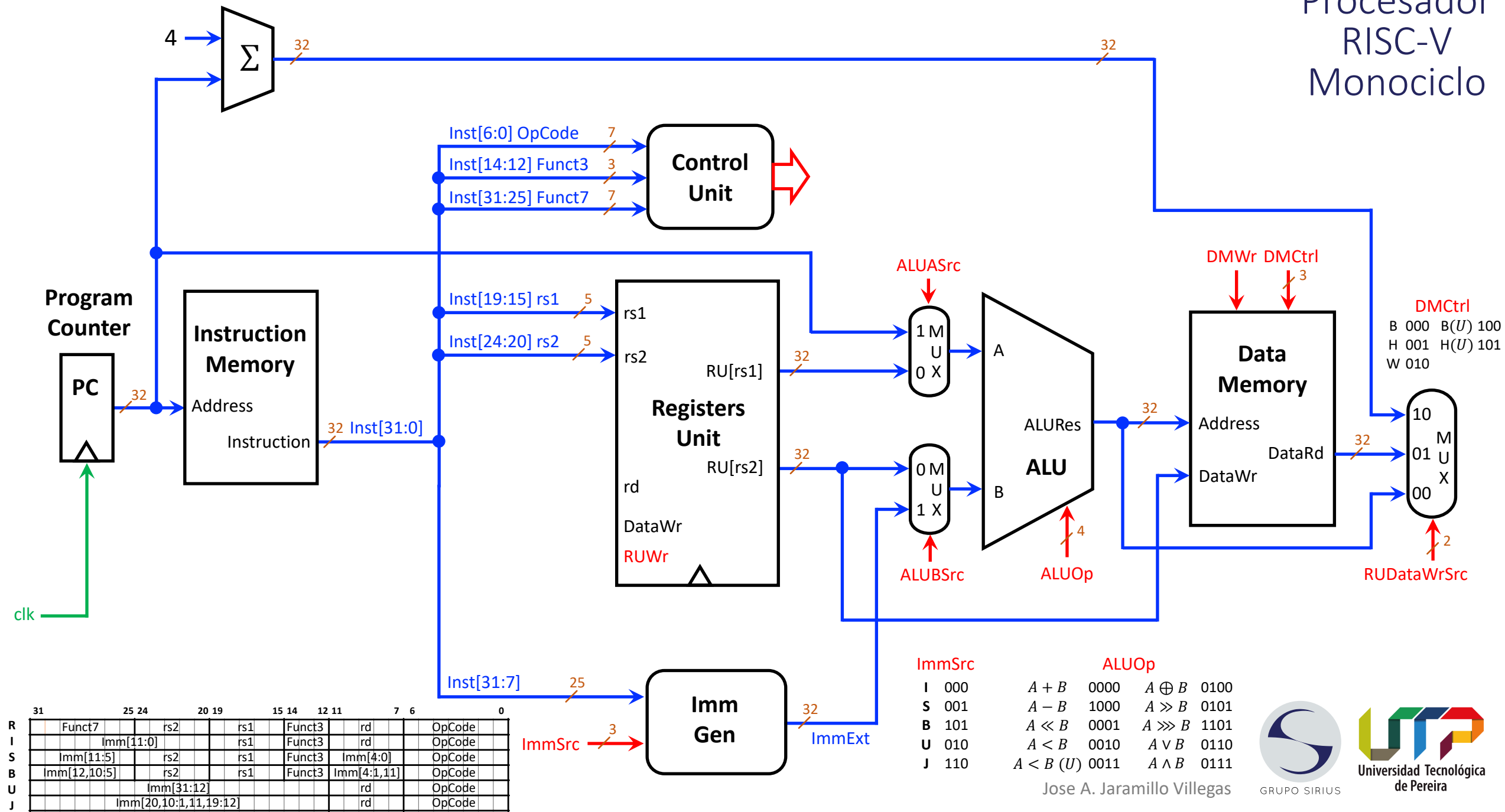




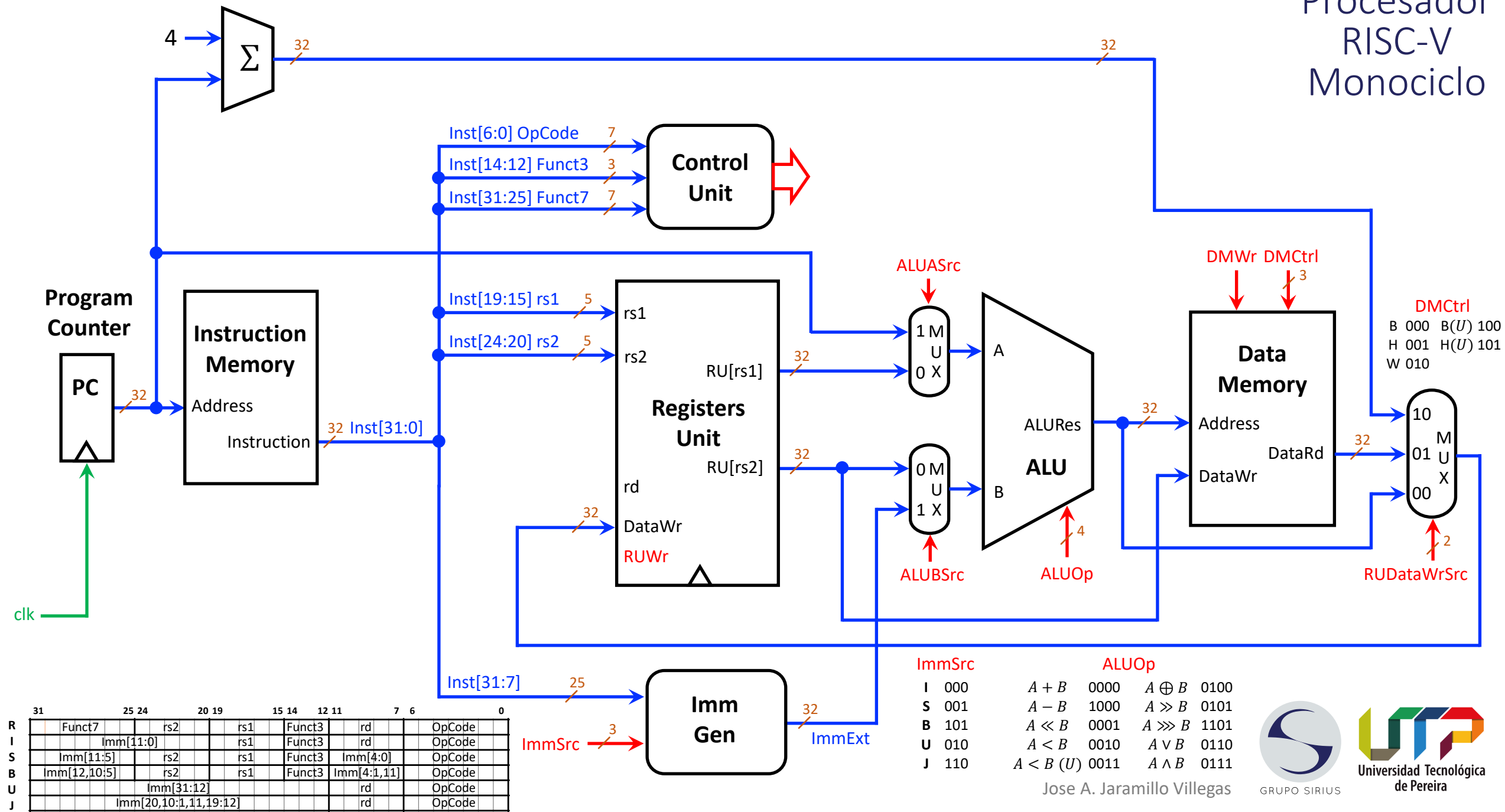
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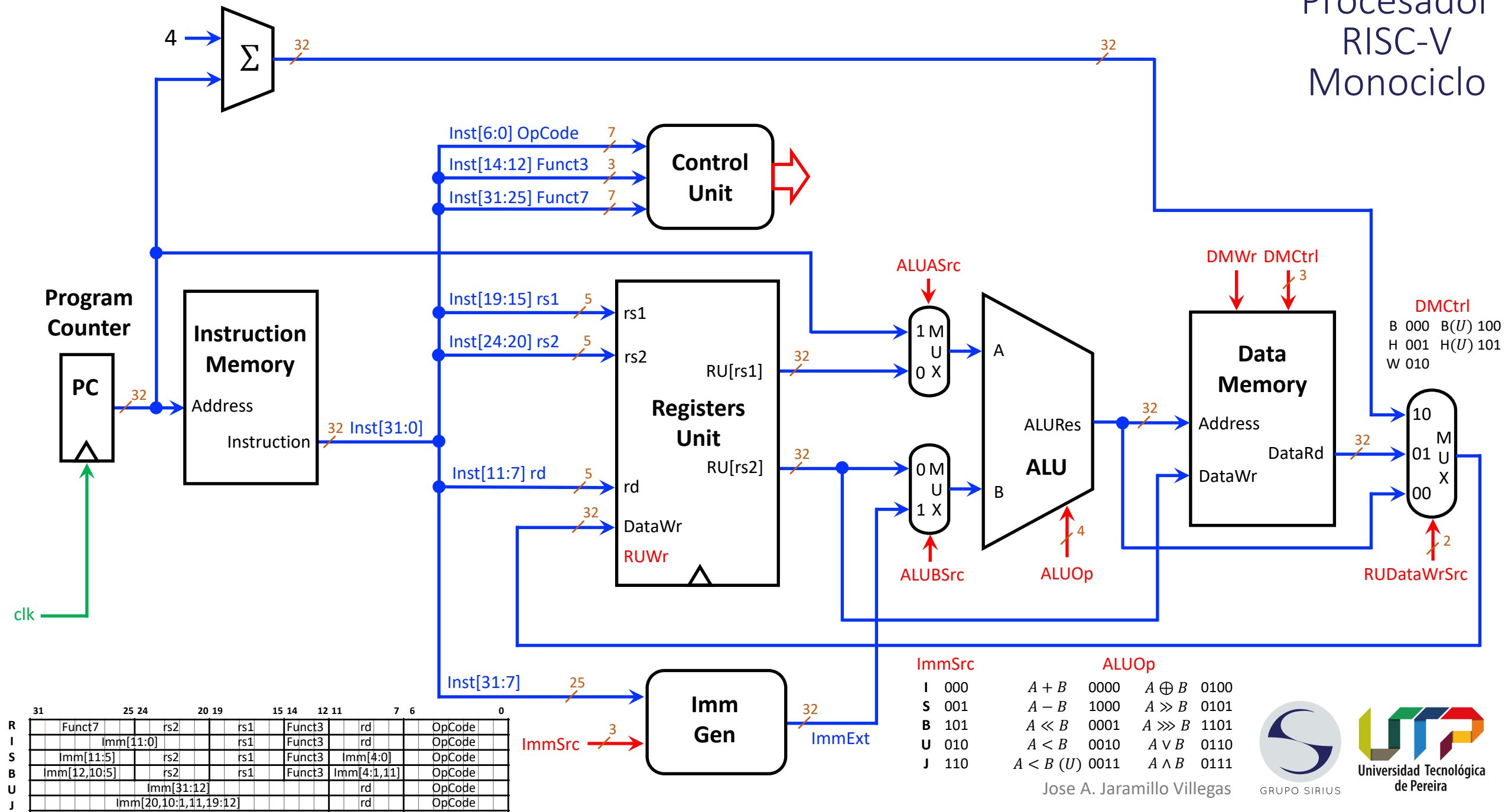
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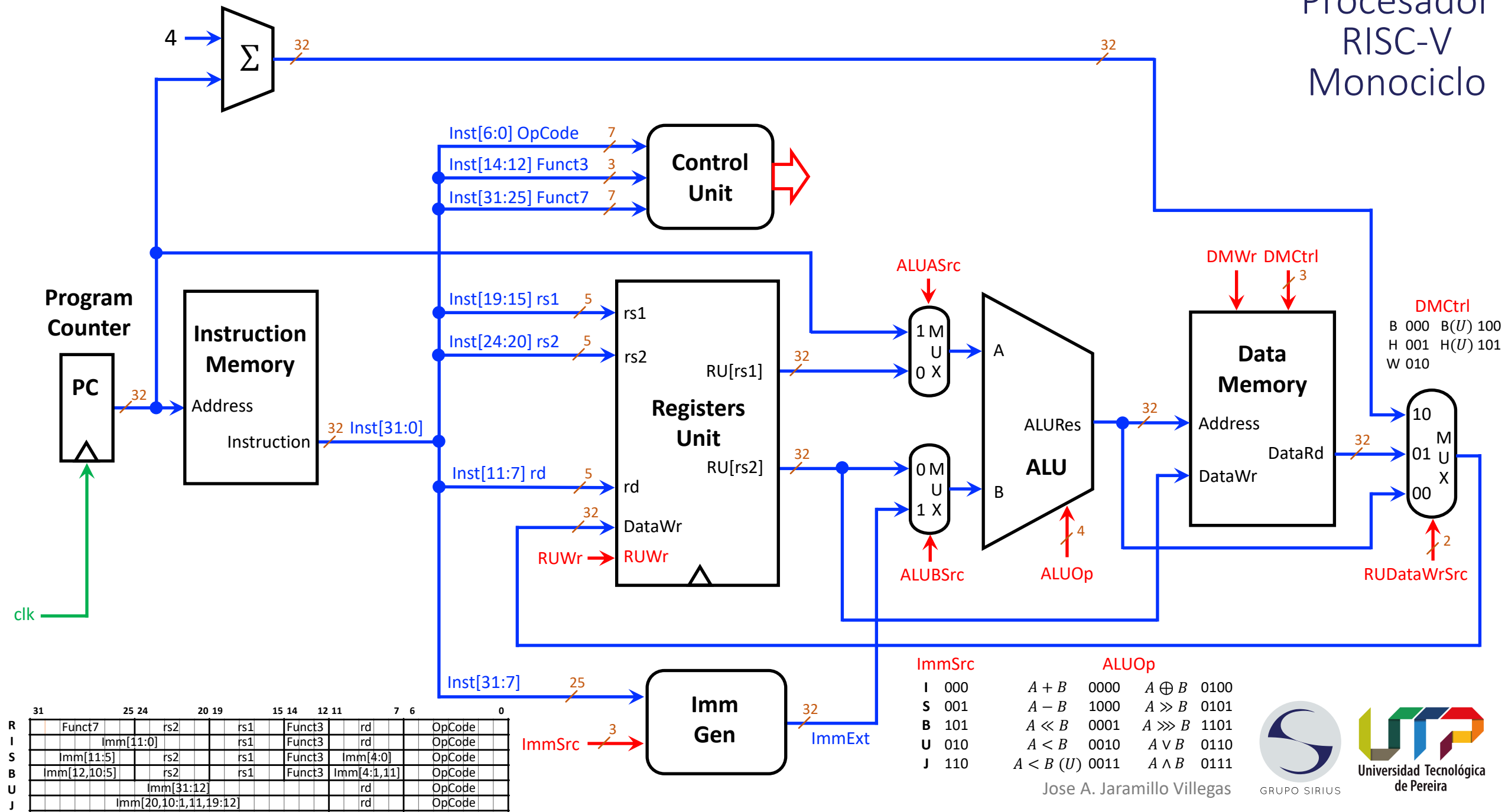
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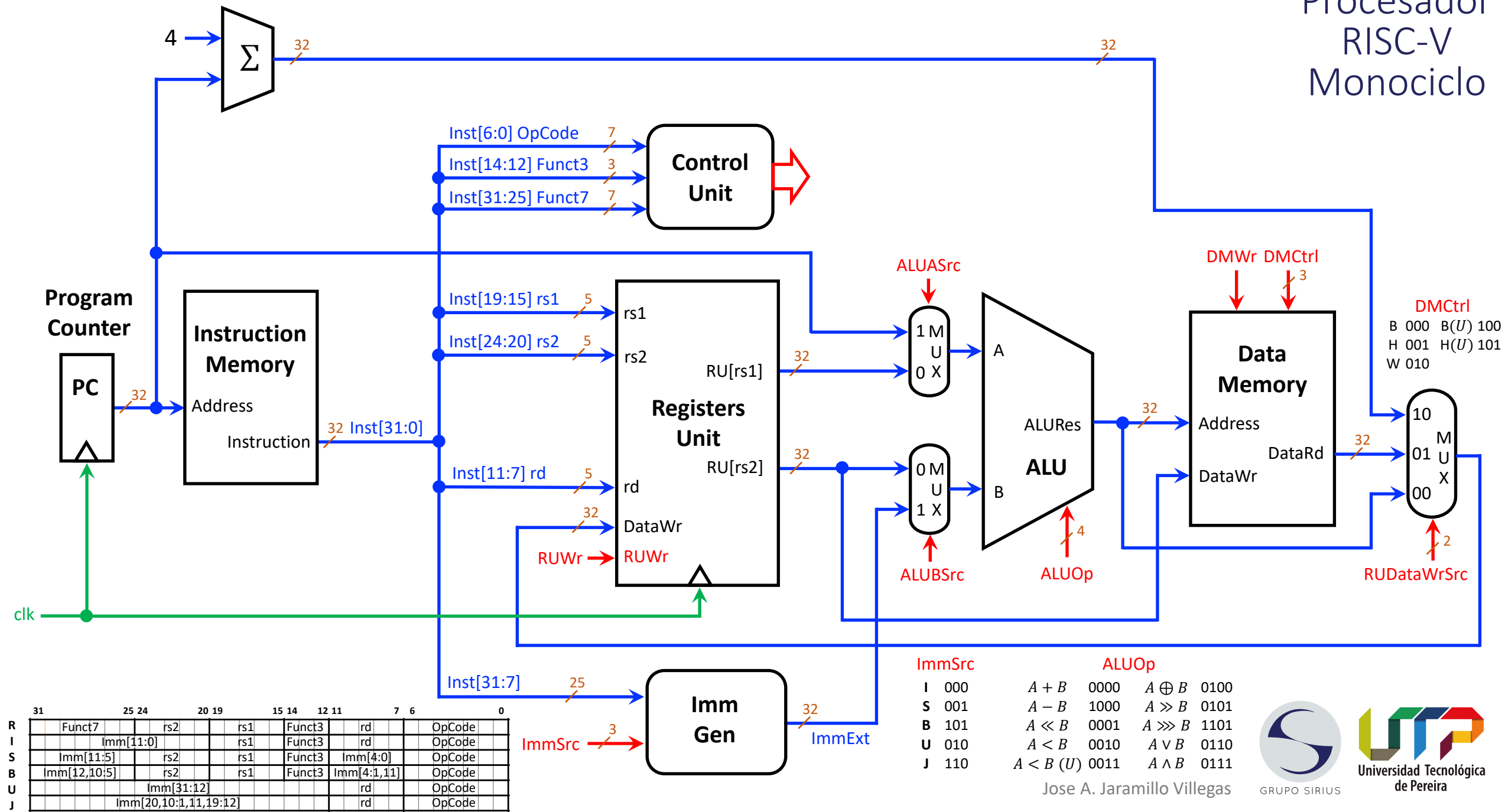
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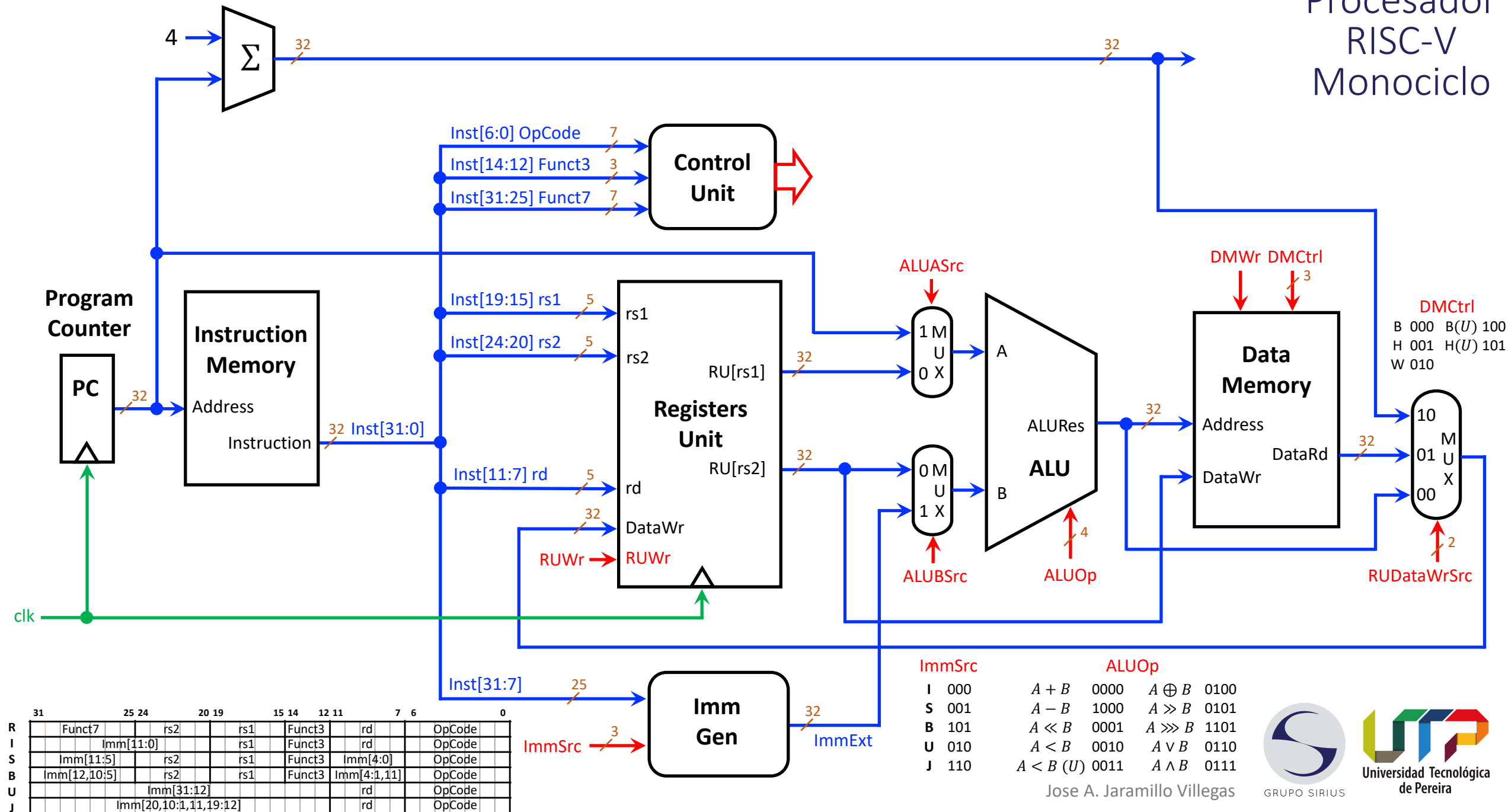
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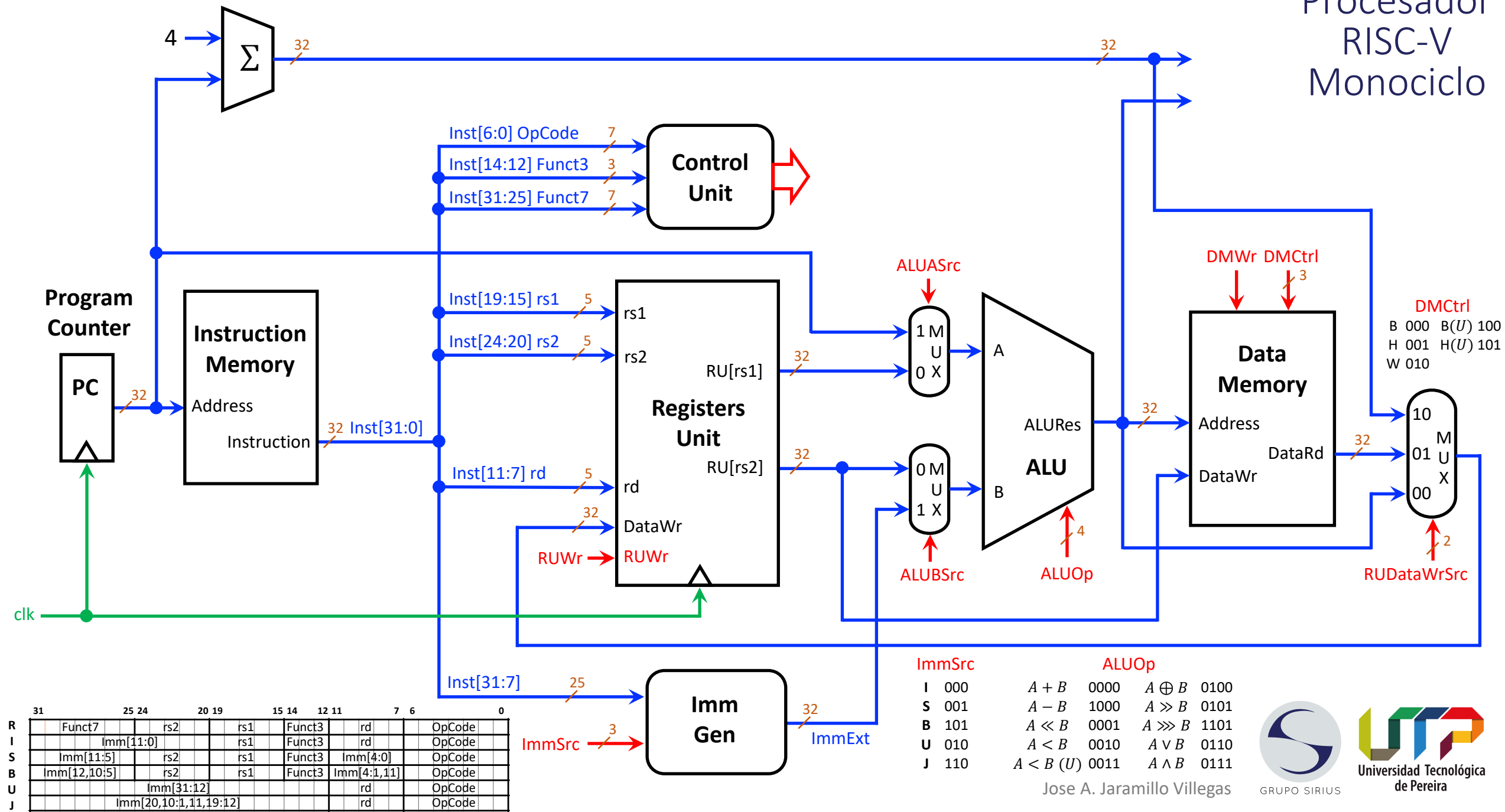
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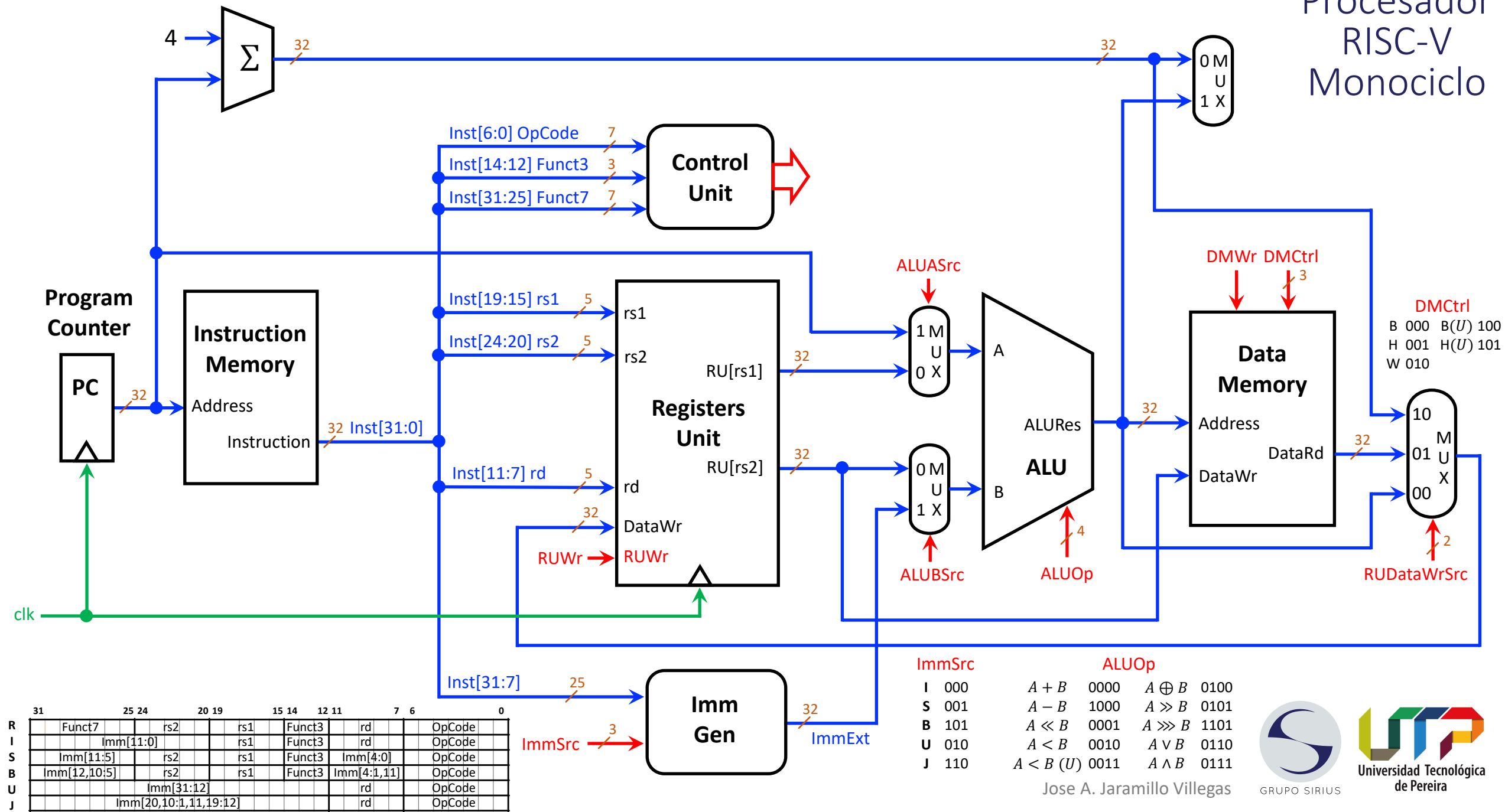
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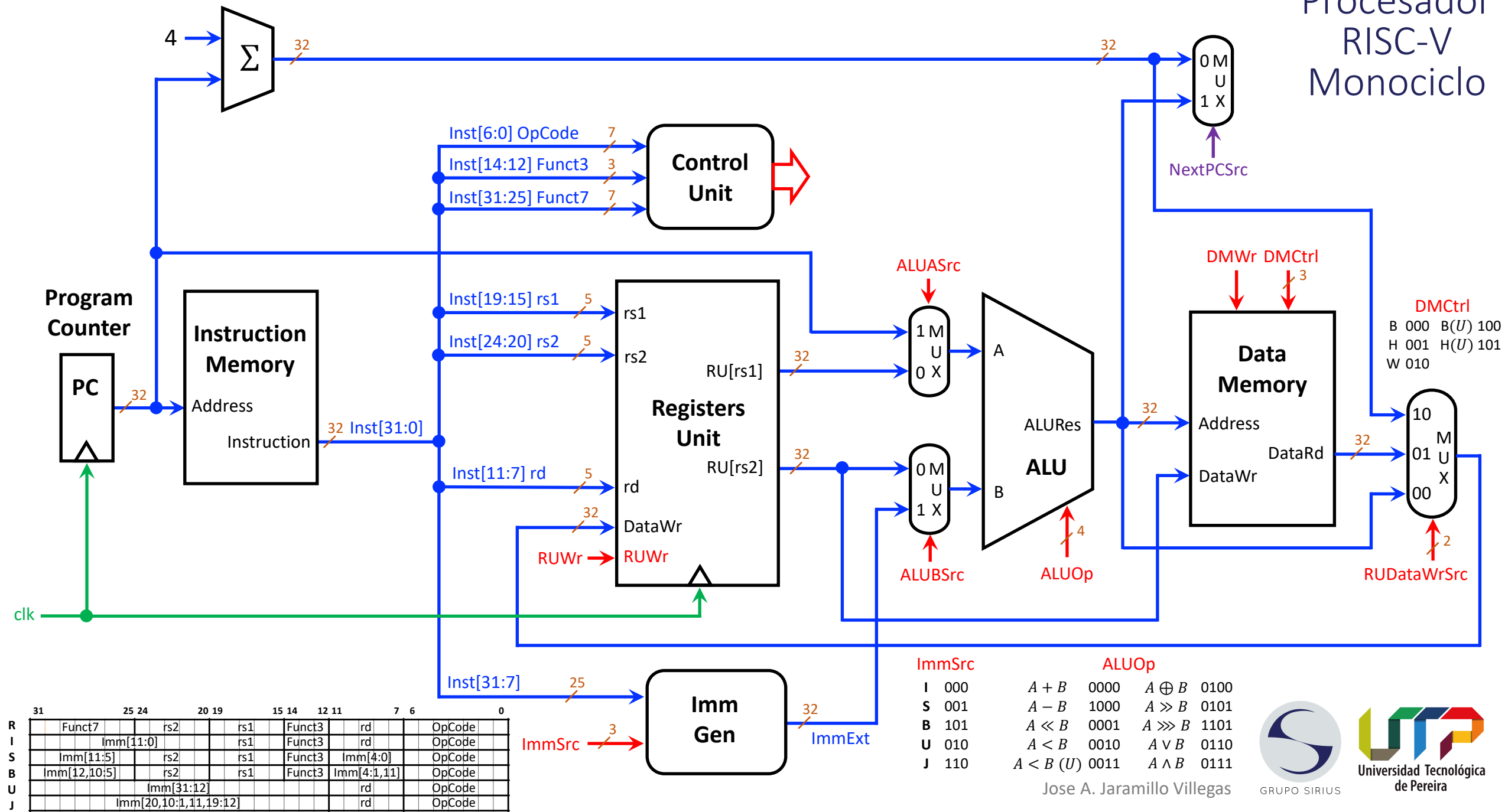
Procesador RISC-V Monociclo



Procesador RISC-V Monociclo

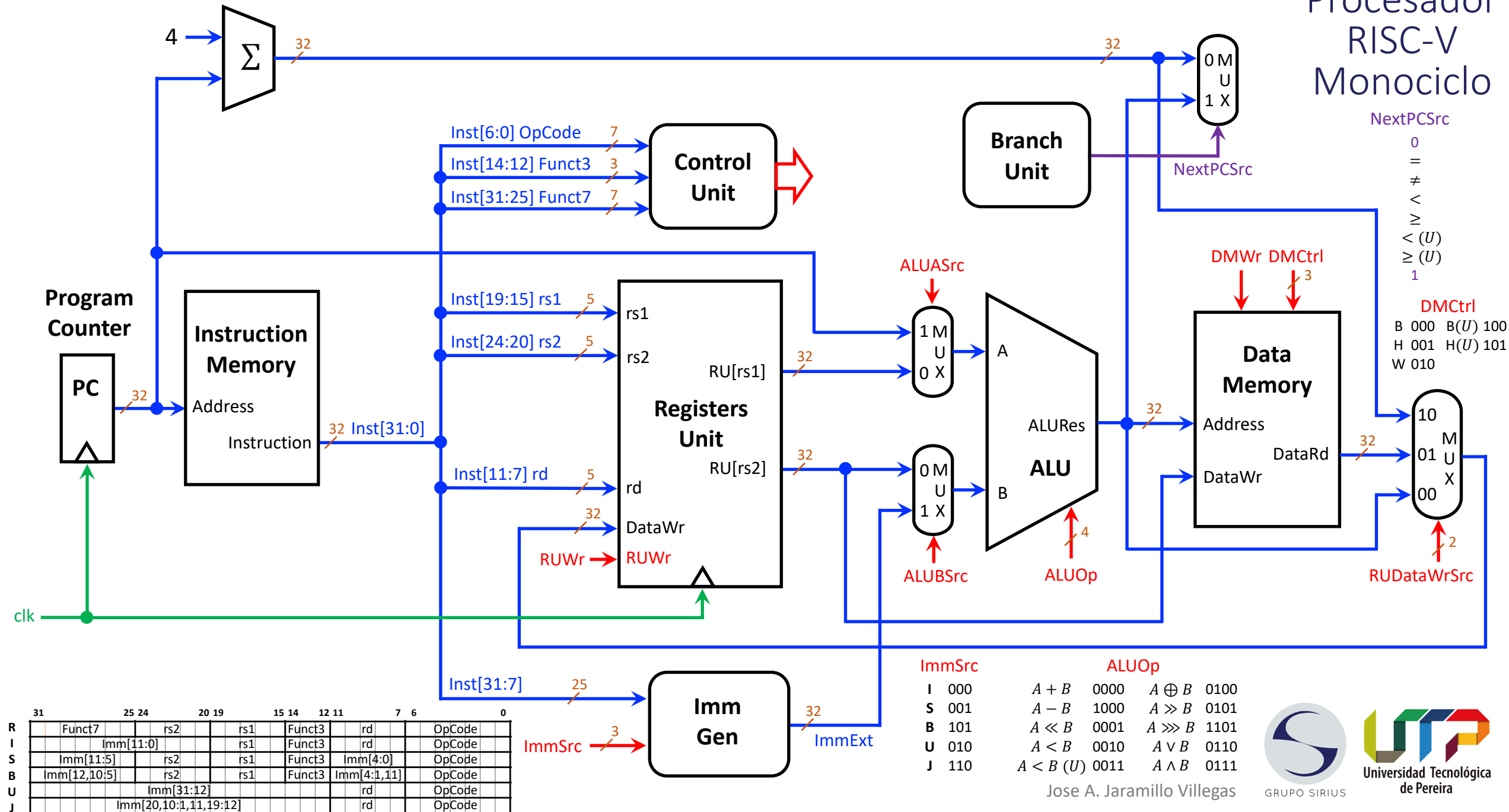


Procesador RISC-V Monociclo

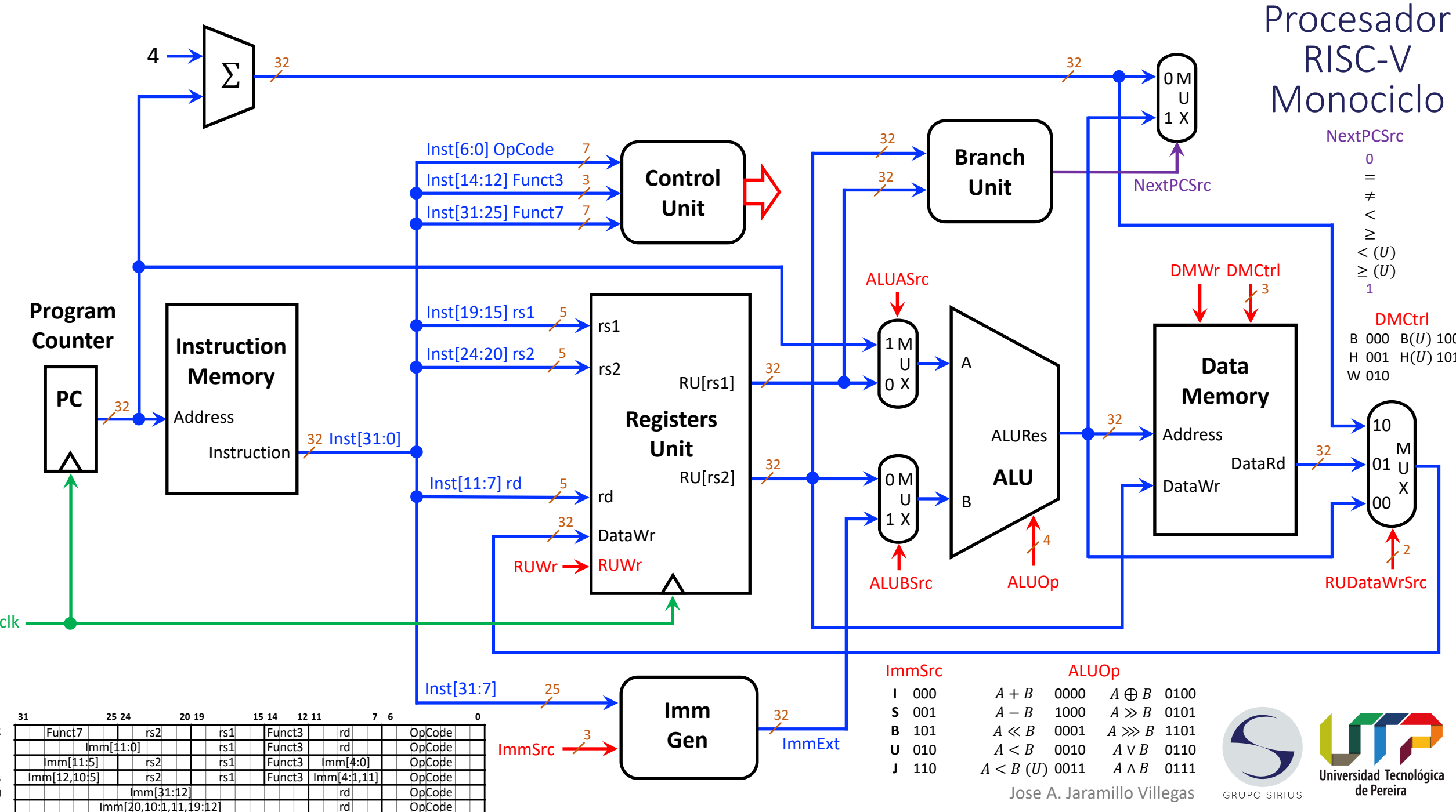




Procesador RISC-V Monociclo



Procesador RISC-V Monociclo



0
 $=$
 $\neq$
 $<$
 $\geq$
 $< (U)$
 $\geq (U)$
 1

DMCtrl
 B 000 B(U) 100
 H 001 H(U) 101
 W 010

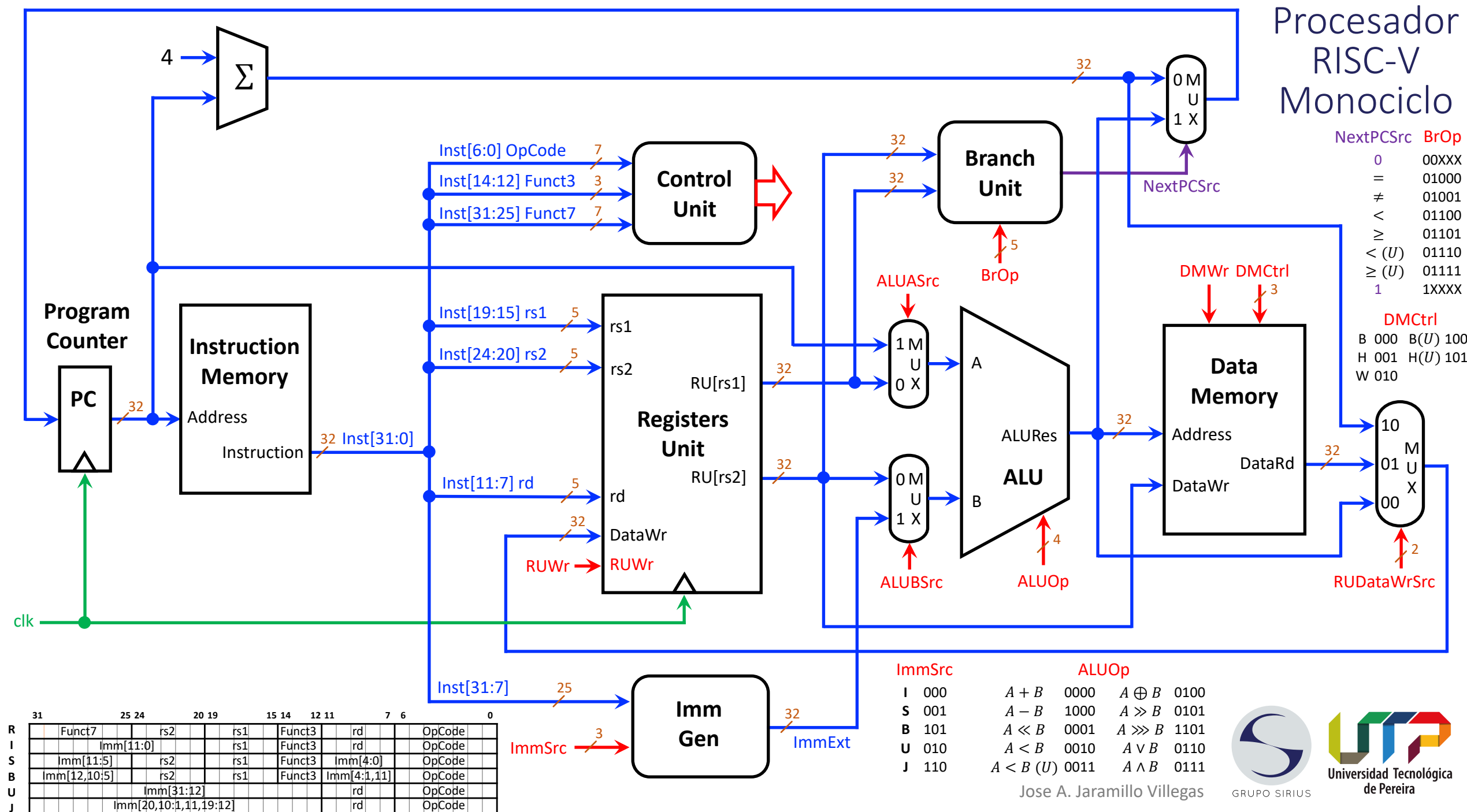


NextPCSrc	BrOp
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=	01000
≠	01001
<	01100
≥	01101
< (<i>U</i>)	01110
≥ (<i>U</i>)	01111
1	1XXXX

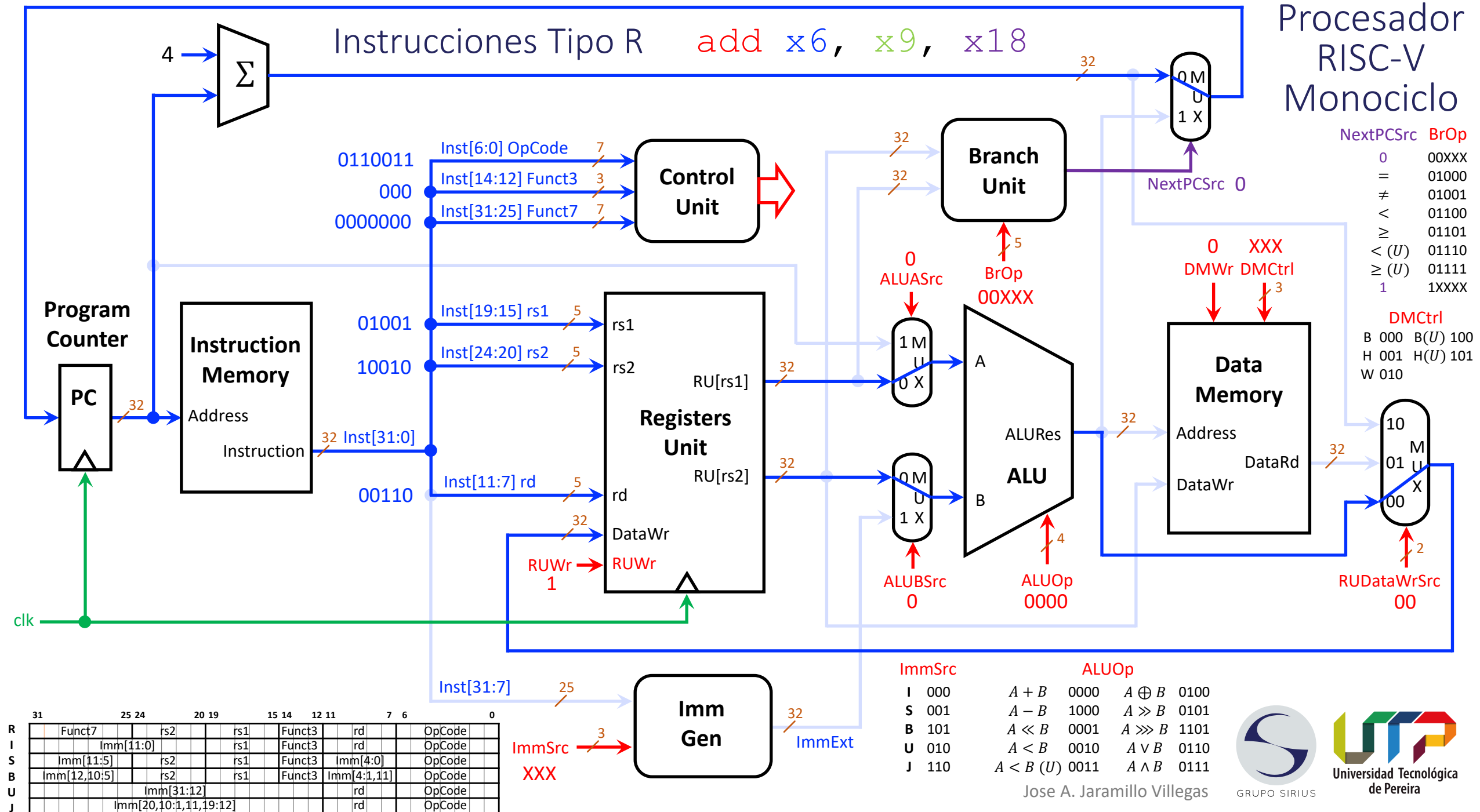


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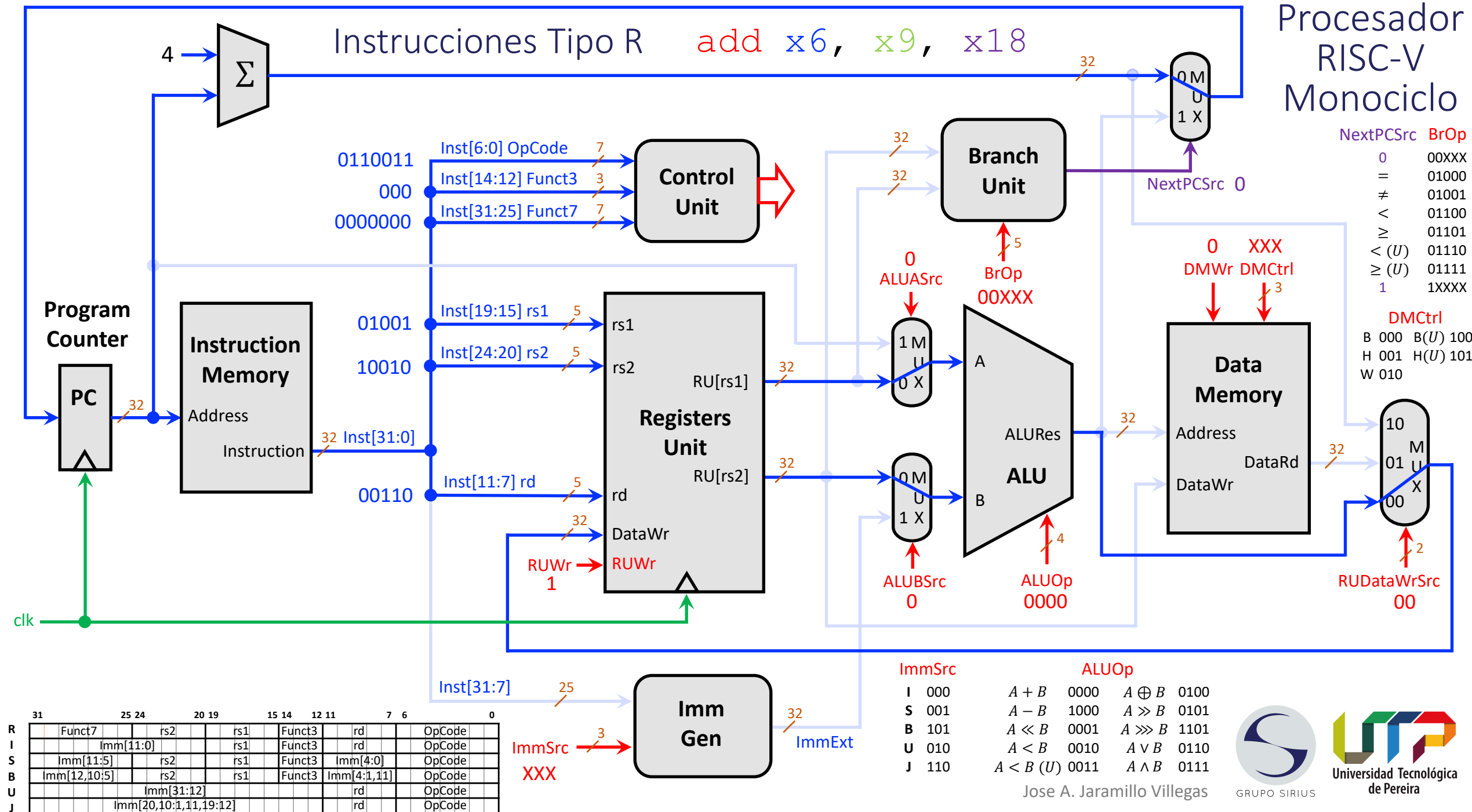
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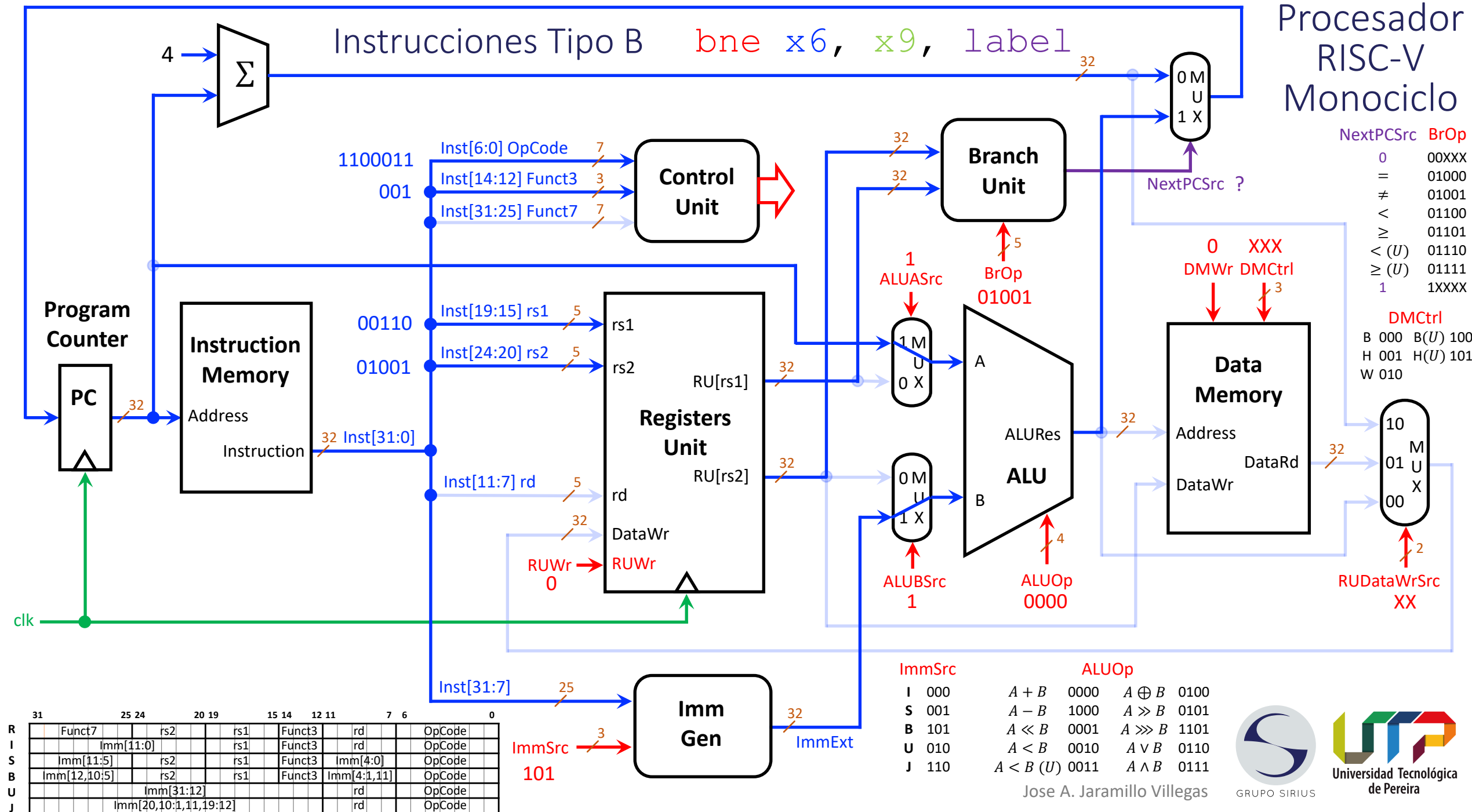
Procesador RISC-V Monociclo



Procesador RISC-V Monociclo



Procesador RISC-V Monociclo



Procesador RISC-V Monociclo

